

HGCAM

Digital Power Meter

HD HYUNDAI ELECTRIC

Essential for Today, Potential for Tomorrow

HD Hyundai Electric solely pursues the growth of our customers' business. From power generation to power distribution, we focus on developing and commercializing products and solutions aimed at increasing the efficiency of energy equipment as well as at proactively monitoring and controlling assets in an integrated manner to improve our customers' productivity and management efficiency. We are well aware that our efforts add to the driving force behind our customers' growth and contribute to the creation and maintenance of a more dynamic world. We focus on achieving innovation and strive to evolve continuously to shape a better tomorrow based on today's technological advancement

Solution

INTEGRICT

Energy Solution

Energy solution business refers to the business of designing, procuring and establishing a system that enables the efficient use of power energy through integrated management of the production, consumption, sales and operation.

Asset Management Solution

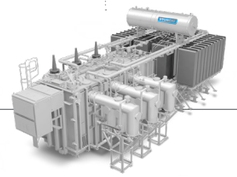
Asset management solution is a business that maximizes the overall business efficiency by systematically managing the performance, risk, maintenance cost and others as well as by providing an asset management solution suitable to the customer's circumstance depending on the product lifecycle (PLC) of various products.

Utility

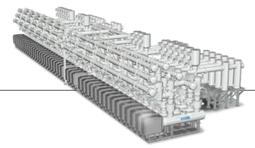
Generation(step-up)

Power Plants

- Supplied to more than 70 countries around the world for the past 45 over years since 1978
- Satisfies the various demands of customers through the acquisition of quality certifications from international accredited institute
- Participates in the world's key technical committee such as GIGRE and others, pioneering the establishment of technology standard related to power network



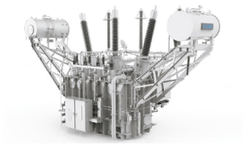
Power Transformer
· up to 800 kV, 1,500 MVA



Gas Insulated Switchgear
· up to 800 kV

- Can be installed in spaces smaller than the open type of substation by using SF_6 gas with outstanding insulation and arc extinguishing characteristics
- Manufacturing advanced products that are resistant to external environment and climate effects through the sealing at the charge part
- Extensive project experiences around the world
- Reduces installation period and cost due to simple installation and transportation, convenient maintenance
- Design considering the safety of the workers as priority

GREENTRIC



ECO Transformer
· up to 500 kV, 1,000 MVA

- Eco-friendly oil (Natural/Synthetic)
- Dynamic Pressure Resistant System(DPRS)
- Guarantees excellence in safety

GREENTRIC



SF_6 Free Gas Insulated Switchgear
· GIS for 170 kV 50 kA 60 Hz
· ECO GIS for 72.5 kV, 145 kV

- Korea's first solely developed technology
- Eco-friendly Novec mixture gas instead of SF_6 gas

Transmission

Primary Substation/Secondary Substation



Gas Insulated Switchgear
· up to 800 kV



Power Transformer
· up to 800 kV, 1,500 MVA



Shunt Reactor
· up to 765 kV, 300 Mvar

Distribution



Cubicle GIS
· up to 40.5 kV
· IEC, ANSI, GOST, UL

- Produces high quality products using angle-less type
- Multi-functional digital protection relay (HiMAP) applied
- High reliability secured, provides various operation information such as protection, measurement and control
- Firm external box, size and compact, making it safe
- Maintains high quality through stringent quality control system and continuous research and development

Motor & Generator



Synchronous Generator
· Rated Output: < 50 MVA
· Rated Voltage: < 15 kV
· Ploes: > 2P

Marine

Marine Solution

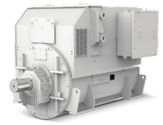
- Production of high quality marine devices satisfying the regulations and standards of key marine associations (LRS, ABS, DNV, GL, BV, NK etc.) and world's renowned institutes
- High quality safety secured through the latest equipment and stringent quality control system
- Realization of optimal high efficiency by converging SWGR, Generator, Motor, Telecom, Automation, Intelligent Motor Control Unit and others



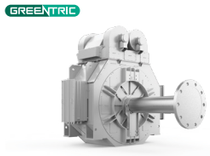
Marine Switchgear



Marine Motor



Generator



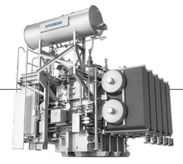
Shaft Generator



Metal Clad Switchgear
· up to 38 kV
· IEC, ANSI, UL



Low Voltage Switchgear & Motor Control Center
· H8PU : 480 V, 3,200 A, 65 kA
· H5600 : 480 V, 1,200 A, 65 kA
· HiMCC : 480 V, 5,000 A, 100 kA



Distribution Transformer
· up to 36 kV, 50 MVA



Cast Resin Transformer
· up to 36 kV, 20 MVA



Synchronous Motor
· Rated Output: < 40 MW
· Rated Voltage: < 15 kV
· Ploes: > 2P



MV&HV Induction Motor
· Rated Output: < 35 MW
· Rated Voltage: < 15 kV
· Poles: 2P-30P



VCB
· IEC, ANSI, UL
· up to 36/38 kV, 50 kA, 4,000 A



ACB
· AC : up to 150 kA, 6,300 A
up to 130 kA, 6,000 A
(UL1066, ANSI C37.13/16/17/50, CSA C22.2 no. 31)
· DC : up to 60 kA, 4,000 A



VC
· up to 12 kV, 400 A



Switch-Disconnector
DC : up to 100 kA/1.0 s, 4,000 A
DC : up to 150 kA/0.6 s, 4,000 A
(UL 489, UL489B)



HGMAP
· HGMAP Series



HGCAM
· HGCAM Series



MCCB
· AC : up to 150 kA, 1,600 A
· DC : up to 100 kA, 800 A



ELCB
· up to 85 kA, 800 A, 1,000 mA



MCCB / ELCB
· up to 10 kA, 100 A, 100 mA



MS / MC
· up to 800 A



Distribution Box
· Up to 36 ways
· Surface/Flush type



MCB
· up to 15 kA, 125 A



RCCB / RCBO
· up to 100 A, 500 mA



Isolator
· up to 125 A



Electronic Circuit Breaker
· up to 20 A



MMS
· up to 100 kA, 80 A



Fuse Links & Switches
· up to 1,250 A

- Can be applied to various places of demand due to various domestic and international certifications and marine certifications
- Full HG-Series line-up established
- High standard of breaking capacity and various product line offers production in any load environments
- Increased reliability with reinforced breaking performance of high-breaking MCCB
- Reinforced usage stability through internal reliability verification



Installation Contactor
· up to 63 A



SPD
· up to 200 kA
· AC, DC



HGDM



EOCR
· up to 60 A

- Enhanced reliability and secured safety with production of products based on the world's best equipment and stringent quality system
- Realized high efficiency by selecting slot based on FEM
- Realized small and lightweight with optimal design based on FEM analysis method
- Satisfies the quality standards of international accredited institutes (IEC, IEEE, CSA, NEMA, API etc.)



LV Motor
· Rated Output: < 750 kW
· Rated Voltage: < 690 V
· Poles: 2P-12P

HG Digital Power Meter

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HGCAM

Selection Table

Digital Power Quality Meter (HGCAM-PQ)

Type		Specification
General		
Application		It measures various amounts of electricity in the distribution system and monitors and controls the system. Power quality monitoring and automatic power factor control can reduce maintenance cost in the system. Each status information can be monitored by remote communication in real time.
Environment	Temperature	Rated Ambient Temperature : 25°C Continuous Operation Temperature : -10~55°C Storage Temperature : -10~75°C
	Humidity	Under 80%, Condensation must be avoided
	Location	Location below 2000M altitude Not exposed to direct sunlight, snow, rain, wind and so on Not exposed to corrosive, explosive gas and dust
Configuration		
Input	Control power	AC/DC 88~265V
	Analog input	PT 3 channel, CT 4 channel
Measure	Voltage	Phase voltage, Line voltage, Positive sequence voltage, Negative sequence voltage, Zero sequence voltage, Voltage unbalance
	Current	Line current, Positive sequence current, Negative sequence current, Zero sequence current, Current unbalance
	Phase	Voltage, Current
	Power	Active power, Reactive power, Apparent power
	Energy	Active energy, Reactive energy, Reverse active energy, Reverse reactive energy, Apparent energy
	Frequency	Frequency
	Power factor	Power factor
	Harmonics	2nd~15th harmonics, voltage, current, voltage THD, current THD/TDD, k-factor
	Demand	Current demand, Power demand
Digital I/O	DI	12 channel
	DO	10 channel
Display	Screen	240x128 graphic LCD
	LED	Status & Alarm LEDs x16
Event log	Event	max. 1024 records
	PQ event	max. 256 records
	Fault wave	max. 10 records (IEEE C37.111 Comtrade format)
Size	Dimension	148x208x188mm (WxHxD)
	Weight	3.5kg (Gross weight)

Type			Specification
Rating			
Control power	Rated power		AC/DC 110/220V
	Input range		AC/DC 88~265V
	Voltage sag immunity		100msec (at 110VDC)
	Consumption		Standby 10W or less, Operation 15W or less
CT input	CT primary set		5~9,000A
	CT secondary set		5A (=1In)
	Maximum load		below 1.0VA at 1In
	Input range		0~24In
	Overload capability		2In(continuous), 20In(for 2sec), 40In(for 50msec)
PT input	PT primary set		110~345,000V
	PT secondary set		100~440V (=1Vn)
	Maximum load		below 0.5VA at 1Vn
	Input range		0~2.0Vn
	Overload capability		1.2Vn(continuous), 2Vn(for 1hour)
Digital Input	Rated input		110VDC
	Threshold voltage		Turn-on 80VDC, Turn-off 70VDC
	Input delay		within 10msec
	Maximum load		0.5VA at 1Vn
Digital Output (control)	Contact type		Dry contact
	Contact rating		Resistive load : 10A at 250VAC/30VDC
			Inductive load : 5A at 250VAC/30VDC
	Operating speed		within 10msec
Digital Output (signal)	Contact type		Dry contact
	Contact rating		Resistive load : 5A at 250VAC/30VDC
			Inductive load : 2A at 250VAC/30VDC
	Operating speed		within 10msec
Communication	RS-485	Wiring	2-wire(D+, D-), 4-wire(Rx+, Rx-, Tx+, Tx-)
		Baudrate	9600, 19200, 38400, 57600 bps
		Protocol	Modbus/RTU
	Bluetooth	Version	4.0
	Manager S/W	Connector	mini USB-B port
		Protocol	Reserved

Selection Table

Digital Power Quality Meter (HGCAM-PQ)

Measurement

Item		Acronym	Range	Unit	Accuracy
Voltage	Phase voltage	Va,Vb,Vc	0.000~999.99	kV	±0.2%
	Zero sequence voltage	Vo	0.000~999.99	kV	±0.2%
	Line voltage	Vab,Vbc,Vca	0.000~999.99	kV	±0.2%
	Positive sequence voltage	V1	0.000~999.99	kV	±1.0%
	Negative sequence voltage	V2	0.000~999.99	kV	±1.0%
	Unbalance rate	Vu	0.00~100.0	%	±1.0%
Current	Phase current	Ia,Ib,Ic	0.000~999.99	kA	±0.2%
	Zero sequence current	Io	0.000~999.99	kA	±0.2%
	Positive sequence current	I1	0.000~999.99	kA	±1.0%
	Negative sequence current	I2	0.000~999.99	kA	±1.0%
	Unbalance rate	Iu	0.00~100.0	%	±1.0%
Phase	Line voltage	∠Vab,∠Vbc,∠Vca	0.0~360.0	°	±0.5°
	Phase current	∠Ia,∠Ib,∠Ic	0.0~360.0	°	±0.5°
Power	Active power	P,Pa,Pb,Pc	0.000~99.99	GW	±0.5%
	Reactive power	Q,Qa,Qb,Qc	0.000~99.99	Gvar	±0.5%
	Apparent power	S,Sa,Sb,Sc	0.000~99.99	GVA	±0.5%
Energy	Forward active energy	+Ep,+Epa,+Epb,+Epc	0.000~99.99	GWh	±0.5%
	Reverse active energy	-Ep,-Epa,-Epb,-Epc	0.000~99.99	GWh	±0.5%
	Forward reactive energy	+Eq,+Eqa,+Eqb,+Eqc	0.000~99.99	GVarh	±0.5%
	Reverse reactive energy	-Eq,-Eqa,-Eqb,-Eqc	0.000~99.99	GVarh	±0.5%
	Apparent energy	Es,Esa,Esb,Esc	0.000~99.99	GVAh	±0.5%
Power factor	Power factor	PF	-1.00~1.00[F/R]	-	±0.5%
	Phase power factor	PFa, PFb, PFc	-1.00~1.00[F/R]	-	±0.5%
Frequency	Frequency	Freq	45.000~75.000	Hz	±0.005Hz
Harmonics	Phase voltage(n=<15) 1)	Va[n],Vb[n],Vc[n]	0.00~999.99	kV	-
	Line voltage(n=<15) 1)	Vab[n],Vbc[n],Vca[n]	0.00~999.99	kV	-
	Phase current(n=<15)	Ia[n],Ib[n],Ic[n]	0.00~999.99	kA	-
	Voltage THD	THD Va,Vb,Vc	0.00~100.00	%	-
	Current THD	THD Ia,Ib,IC	0.00~100.00	%	-
	TDD	TDD Ia,Ib,Ic	0.00~100.00	%	-
	k-factor	Ia,Ib,Ic	-	-	-
Demand	Current demand	Demand Ia,Ib,Ic	0.000~999.99	kA	-
	Active power demand	Demand P	0.000~999.99	MW	-
	Reactive power demand	Demand Q	0.000~999.99	MVar	-

※ 1) It depends on PT secondary wiring setting. (WYE: phase voltage, DELTA: line voltage)

Digital Power Meter (HGCAM-A)

Environment

- Storage temperature / Operation temperature : -10°C ~ 55°C / -25°C ~ 70°C
- Humidity : Under 80%, Condensation must be avoided
- Location : Below 2,000M altitude, Not exposed to abnormal vibration or shock

Rating - Input

Type	Specification	Note
Wiring	3P4W, 3P3W(Wye), 3P3W(Delta)	
Frequency	60Hz, 50Hz	
Rated voltage	AC 100~440V	
Voltage range	AC 10~440V	Secondary voltage
Rated current	AC 5A	
Current range	AC 0.05~6A	Secondary current
Burden	1VA or less	VT, CT
DI power	dry, wet	
DI contacts	10 channels for general, 2 channels for control	
Control power	AC/DC 90~265V	
Power consumption	10W or less(standby), 30W or less(operation)	

Rating - Output

Type	Specification	Note
DO power	AC/DC 110V or 220V	
DO contacts	3 channels for signal, 2 channels for control	
DO capacity	AC 240V 16A / DC 30V 16A	
Max. capacity	3,840VA, 480W	

Applied Standard

Test	Class	Standard
Measurement accuracy	Class 0.5S	IEC 62053-22
Electrostatic Discharge(ESD)	8/15kV Contact/Air	IEC 61000-4-2
Fast Transient Burst	4kV	IEC 61000-4-4
Surge Immunity	4kV	IEC 61000-4-5
Conducted RF Immunity	10V	IEC 61000-4-6
Voltage Dip/Shot Interrupts	1cycle, each polarity 100%	IEC 61000-4-11
Cold Test	-25°C, 72hour	IEC 60068-2-1
Dry Heat Test	70°C, 72hour	IEC 60068-2-2
Vibration Test	10~150Hz, 1g, 3pole	IEC 60068-2-6
Shock Test	30gn	IEC 60068-2-27
Damp Heat Cycle Test	40°C indoor, 144hour	IEC 60068-2-30

Selection Table

Digital Power Meter (HGCAM-A)

Measurement

Item		Acronym	Range	Unit	Accuracy
Voltage	Phase voltage	Va,Vb,Vc	<1k : 0.00~999.99 <100k : 1000.00~99999.9 >100k : 100.000~99999.9k	V/kV	±0.3%(F/S)
	Line voltage	Vab,Vbc,Vca			
	Zero phase voltage	Vo			
	Positive sequence	V1			-
	Negative sequence	V2			
	Zero sequence	V0			
	Unbalance rate	Vu	0.00~100.0	%	-
Current	Phase current	Ia,Ib,Ic	<1k : 0.00~999.99 <100k : 1.00~999.99 >100k : 1000.00~99999.9	A	±0.3%
	Zero phase current	Io			
	Positive sequence	I1			-
	Negative sequence	I2			
	Zero sequence	I0			
	Unbalance rate	Iu	0.00~99.9	%	-
	Phase current load rate	LIa,LIb,LIc	0.00~120.0	%	-
Phase	Voltage	∠V	0.0~359.9	°	±0.5°
	Current	∠I			
Power	Active power	P,Pa,Pb,Pc	<1k:0.000~0.999 <1000k:1.00~999.99 <10G:1000.0~9999999.9 >10G:10000000~99999999	kW/MW	±0.5%
	Reactive power	Q,Qa,Qb,Qc		kVar/MVar	
	Apparent power	S,Sa,Sb,Sc		kVA/MVA	
	Power load rate	Lp	0.0~250.0	%	
	Phase power load rate	Lpa,Lpb,Lpc			
Energy	Forward active energy	+Ep,+Epa,+Epb,+Epc	<1k:0.000~0.999 <1000k:1.00~999.99 <10G:1000.0~9999999.9 >10G:10000000~99999999	kWh/MWh	
	Reverse active energy	-Ep,-Epa,-Epb,-Epc			
	Forward reactive energy	+Eq,+Eqa,+Eqb,+Eqc		kVarh/Mvarh	
	Reverse reactive energy	-Eq,-Eqa,-Eqb,-Eqc			
	Apparent energy	Es,Esa,Esb,Esc		kVAh/MVAh	
Power factor	Power factor	PF	-1.000~+1.000[F/R]	-	-
	Phase power factor	PFa, PFb, PFc			
Frequency	Frequency	Freq	45.00~75.00	Hz	±0.005Hz
Demand	Current demand	Demand Ia,Ib,Ic	0.00~99999.9	A	-
	Active power demand	Demand P	0.000~9999999.9	kW/MW	-
Max	Max line voltage	Va,Vb,Vc	0.00~99999.9	V/kV	±0.3%
	Max current	Ia,Ib,Ic		A	
	Max zero phase voltage	Vo		V/kV	
	Max zero phase current	Io		A	
	Max active power	P	0.000~99999999	kW/MW	±0.5%
	Max reactive power	Q		kVar/MVar	
	Max apparent power	S		kVA/MVA	
Harmonics	Voltage THD	THD Va,Vb,Vc	0.00~100.00	%	-
	Current THD	THD Ia,Ib,Ic			
	TDD	TDD Ia,Ib,Ic			
	k-factor	Ia,Ib,Ic	-	-	-
	Phase voltage(n=<15)	Va[n],Vb[n],Vc[n]	0.00~99999.9	V/kV	-
	Phase current(n=<15)	Ia[n],Ib[n],Ic[n]	0.00~99999.9	A	-

Digital Power Meter (HGCAM-S)

Input/Output

Type		Channel	Capacity	Note
Input	Voltage	3	-	Default
	Current	3	-	Default
	DI Contact	3	-	Option(DI1~3)
Output	DO Contact	2	AC220V/5A	Option(DO1~2), aux. relay for CB control

General Specification

Type		Specification
Wiring		1P2W, 1P3W, 3P3W, 3P4W
Rating	Current	5A(0.05~6A), burden 0.2VA
	Voltage	0~380V(10~440V), burden 0.1VA(220V, line)
	Frequency	50Hz, 60Hz
Measurement	Accuracy	Voltage/Current ±0.3%, Power/Energy 0.5class
Temperature	Operation	-10~55°C
	Storage	-25~75°C
Overload withstand	Current circuit	Withstand 3 times of rated current for 3 hours, withstand 20 times of rated current for 1 second
	Voltage circuit	Withstand 3 times of rated voltage for 3 hours
	Aux. power	Withstand 1.3 times of rated voltage for 3 hours
Communication	Protocol	Modbus-RTU Protocol (9600/19200/38400/57600bps)
	Media	RS-485/Multi drop
Control power		AC/DC 90~265V
Power consumption		2W or less(standby)
Dimension		127(W)x127(H)x115.8(D)mm
Weight		0.8kg

Applied Standard

Test	Class	Standard
Measurement accuracy	Class 0.5S	IEC 62053-22
Electrostatic Discharge(ESD)	8/15kV Contact/Air	IEC 61000-4-2
Fast Transient Burst	4kV	IEC 61000-4-4
Surge Immunity	4kV	IEC 61000-4-5
Conducted RF Immunity	10V	IEC 61000-4-6
Voltage Dip/Shot Interrupts	1cycle, each polarity 100%	IEC 61000-4-11
Cold Test	-25°C, 72hour	IEC 60068-2-1
Dry Heat Test	70°C, 72hour	IEC 60068-2-2
Vibration Test	10~150Hz, 1g, 3pole	IEC 60068-2-6
Shock Test	30gn	IEC 60068-2-27
Damp Heat Cycle Test	40°C indoor, 144hour	IEC 60068-2-30

Selection Table

Digital Power Meter (HGCAM-S)

Measurment

Item	Range	Scale	Note
Phase voltage	10~440V	0~9,999V,kV	
Line voltage	10~660V	0~9,999V,kV	
Current	0.05~6A	0~9,999A	
Active power	-	0~99,999(x1000)	
Reactive power	-	0~99,999(x1000)	
Apparent power(kVA)	-	0~99,999(x1000)	
Active energy(kWh)	-	0~99,999(x1000)	
Reactive energy(kVarh)	-	0~99,999(x1000)	
Power factor	±1.0	±100.0	
Frequency	-	45.0~65.0Hz	

Accuracy

Item		Acronym	Accuracy
Voltage	Line voltage	Vab,Vbc,Vca	±0.3%
	Phase voltage	Va,Vb,Vc	±0.3%
Current	Current	Ia,Ib,Ic	±0.3%
	load rate	average load rate	LED indicate
Power	Active power	kW	0.5class
	Reactive power	kVar	0.5class
	Apparent power	kVA	0.5class
Energy	Active energy	kWh	0.5class
	Reactive energy	kVarh	0.5class
	Reverse actrive power	rkWh	0.5class
Frequency		Hz	±0.02Hz
Power factor		PF	Depends on phase accuracy

Digital Power DC Meter (HGCAM-S-DC)

Rating

Type		Rating
Frequency		50Hz, 60Hz
AC Voltage	Rating	AC 0~380V
	Range	AC 20~452V
DC Voltage	Rating	DC 0~220V
	Range	DC 20~264V
DC Current	Shunt rating	DC 100mV
	Shunt range	DC 3~250mV
Control power	Rating	AC 110~220V, DC125/220V
	Range	AC/DC 90~265V

Accuracy

Type		Acronym	Accuracy
AC Voltage	Phase voltage	V AC	±0.3%
DC Voltage	Phase voltage	V DC	±0.3%
DC Current	Output current	I DC OUT	±0.5%
	Battery current	I Batt	±0.5%
Frequency		Hz	±0.05Hz

HG Series

Digital Power Quality Meter

HGCAM-PQ

Best Solution for Power Quality Management of System



Overview

Power Quality Management

It monitors various power quality factors as Sag, Swell, Interruption and records them to make use of system management and analysis usefully.

Convenient HMI

The Graphical LCD allows you to view three-phase current/voltage at the same time, and the vector diagram function makes it easy to analyze three-phase deviations. A measurement shortcut button is provided to check the measured values quickly.

Real-time Communication

It supports Modbus/RTU, major industrial communication protocol by RS485 Multi-drop to check the measured value and monitor the system remotely.

APFC(Automatic Power Factor Control)

By controlling the input of an external capacitor bank and stably managing the power factor of the system, facility operating costs can be reduced.

Self-diagnosis

The self-diagnosis function continuously monitors the condition of the device and power system. And when the error occurs, it is notified to user.

Mobile Management App

User can check the measured data and event records safely and easily using smartphone manage application by connecting your personal smartphone through Bluetooth communication without a separate PC or communication equipment.

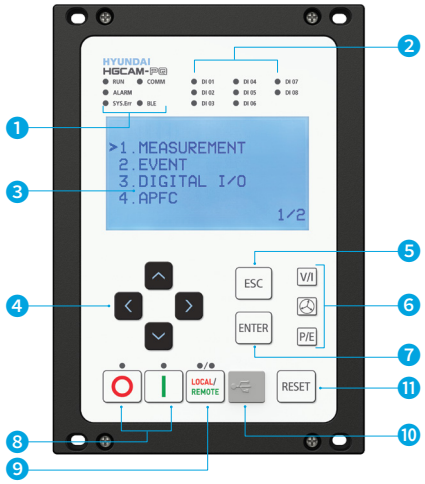


Appearance

Digital Power DC Meter (HGCAM-PQ)

Front View

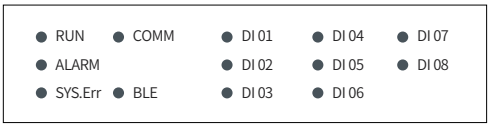
No.	Classification	Description
1	Status LED	Display the status of device and comm.
2	Alarm LED	Alarm for each DI operation
3	Graphic LCD	Display measurement, setting and record
4	Arrow keys	Move between pages, change setting value
5	ESC button	Return to previous page, cancel save
6	Measurement shortcut buttons	Shortcuts for each measurement (voltage/current, vector diagram, power/energy)
7	ENTER button	Enter menu page, confirm save
8	Breaker control button	Control the breaker open/cloase
9	Local/Remote button	Local/remote control to toggle permission
10	USB connector port	mini-B USB port for connection with maintenance laptop
11	RESET button	Cancel the alarm, LED and DO reset control



LED Status Guide

Classification		LED Status	Description	Note
Status LED	RUN	●	Normal operation	
	Alarm	◐ Blinking	Pick-up for PQ event	
		●	Switching of capacitor banks by APFC control	
	SYS.Err	●	System error	
	COMM	◐ Blinking	communication	
	BLE	●	Bluetooth connected	
DI operation LED	DI 01	●	DI 01 state	User-define
	DI 02	●	DI 02 state	User-define
	DI 03	●	DI 03 state	User-define
	DI 04	●	DI 04 state	User-define
	DI 05	●	DI 05 state	User-define
	DI 06	●	DI 06 state	User-define
	DI 07	●	DI 07 state	User-define
	DI 08	●	DI 08 state	User-define

※ LED DISPLAY (default)

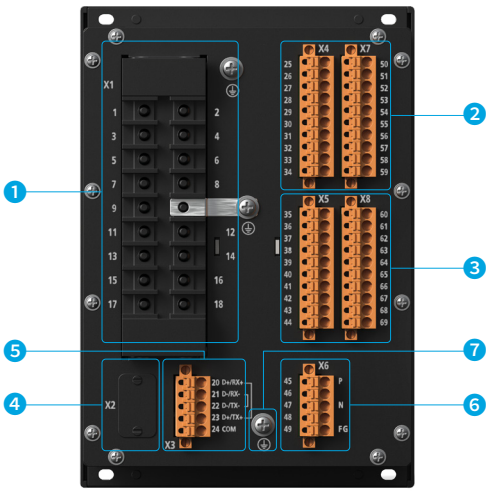


Classification		LED Status	Description
Status LED	O	●	Breaker OPEN
	I	●	Breaker CLOSE
	LOCAL	●	LOCAL control established
	REMOTE	●	REMOTE control established



Rear View

No.	Classification	Description
1	PT and CT input terminal	1~6 : PT terminal (PT1~PT3)
		11~18 : CT terminal (CT1~CT4)
2	DO output terminal	X4:25~28 : CB Close, CB Open
		X4:29~34 : DO01~DO03
		X7:50~56 : DO04~DO08
		X7:58~59 : TCS+, TCS-
3	DI input terminal	X5:35~37 : DI_52a, DI_52b
		X5:38~43 : DI01~DI04
		X8:60~68 : DI05~DI10
4	(spare)	-
5	Modbus comm. terminal	X3:20~24 : RX+, RX-, TX-, TX+, COM
6	Control power terminal	Control power (P, N)
7	ground terminal	Ground connection



Contact Configuration

X1			
1	PT1	2	PT1
3	PT2	4	PT2
5	PT3	6	PT3
7	-	8	-
9	-	10	F.G
11	CT1	12	CT1
13	CT2	14	CT2
15	CT3	16	CT3
17	CT4	18	CT4

X4	
25	CB CLS+
26	CB CLS-
27	CB OPN+
28	CB OPN-
29	DO COM0
30	DO COM0
31	DO COM0
32	DO 01
33	DO 02
34	DO 03

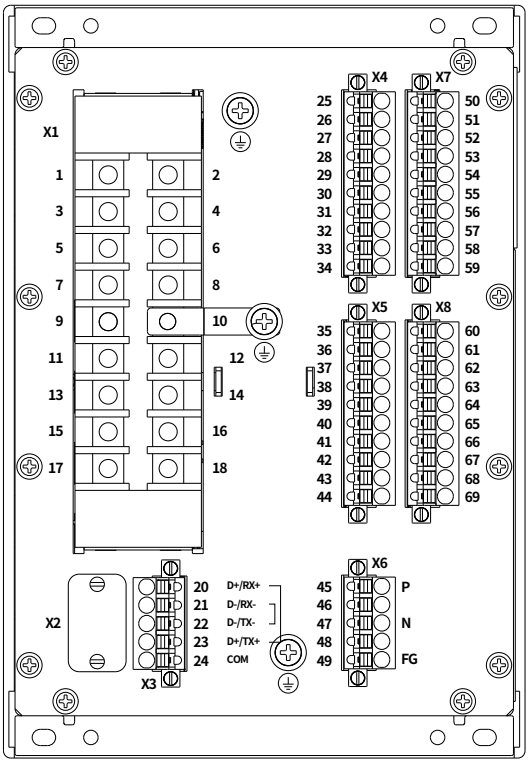
X7	
50	DO COM1
51	DO COM1
52	DO 04
53	DO 05
54	DO 06
55	DO 07
56	DO 08
57	N.C
58	TCS +
59	TCS -

X5	
35	DI 52a
36	DI 52b
37	DI COM0
38	DI 01
39	DI 02
40	DI COM1
41	DI 03
42	DI 04
43	DI COM1
44	F.G

X8	
60	DI 05
61	DI 06
62	DI COM2
63	DI 07
64	DI 08
65	DI COM2
66	DI 09
67	DI 10
68	DI COM2
69	F.G

X2	X3	
	20	D+ / RX+
	21	D- / RX-
	22	D- / TX-
	23	D+ / TX+
	24	COM

X6	
45	P
46	N.C
47	N
48	N.C
49	F.G



Technical Data

Features

Power Quality Management

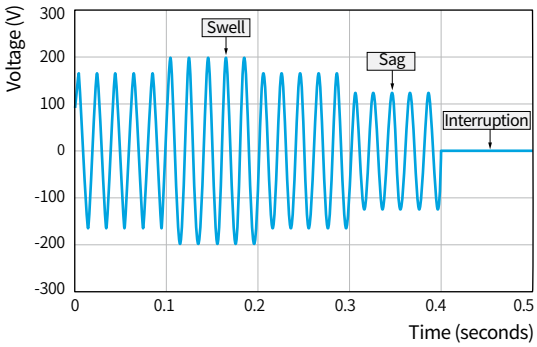
It monitors various power quality factors as Sag, Swell, Interruption and records them to make use of system management and analysis usefully.

Sag (Voltage dip)

- Voltage condition in which the effective voltage is 10% or more and 90% or less of the rated voltage for a certain period of time.
- It is mainly caused by natural disaster such as lightning strikes.

Swell

- Voltage condition in which the effective voltage is at least 110% of the rated voltage and below a certain level for a certain period of time.
- It is caused by ground fault, switching of large-capacity loads or capacitor banks.



Interruption

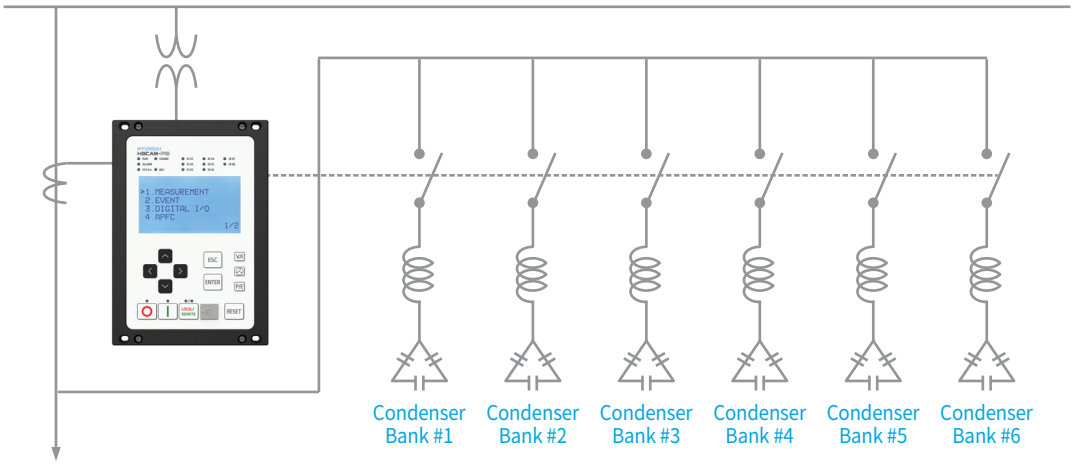
- Voltage condition in which the effective voltage remains below 10% of the rated voltage for a certain period of time.
- It is caused by natural disaster such as lightning, and equipment failures within the distribution system.

Various Event Records

- Internal non-volatile memory is configured to support recording of various elements.
- General events such as product status/setting changes and monitoring are recorded up to 1024.
- Up to 256 events for the operation of power quality monitoring elements are recorded.
- Up to 10 waveform are recorded for the fault event.

APFC(Automatic Power Factor Control)

By controlling the input of an external capacitor bank and stably managing the power factor of the system, facility operating costs can be reduced.

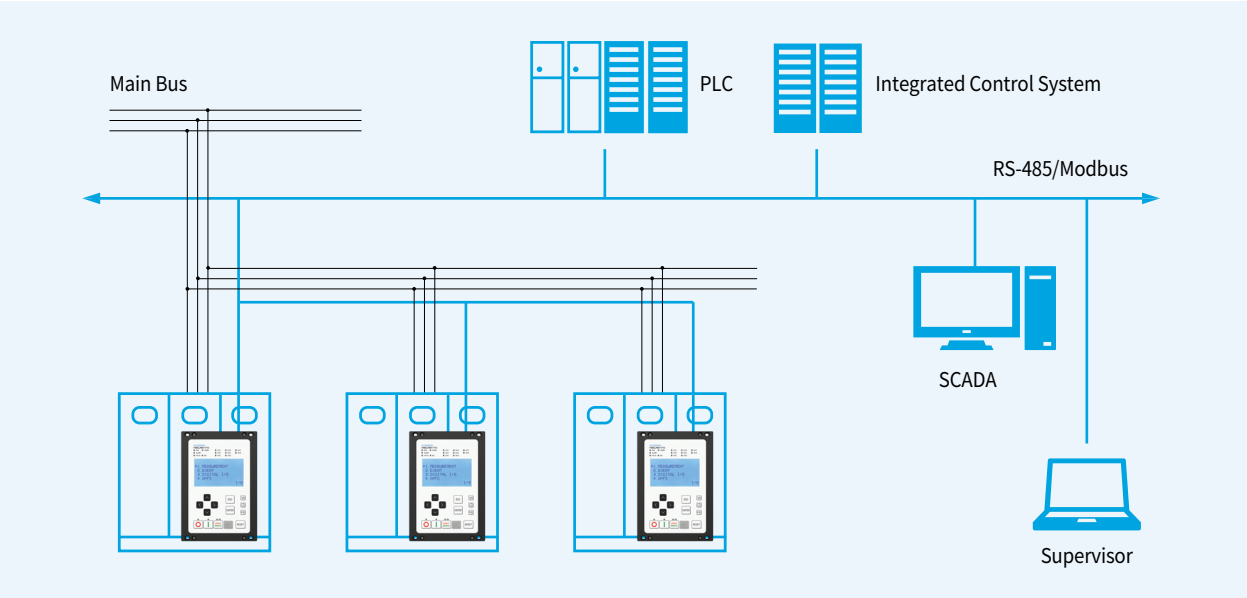


Self-diagnosis

- Various self-diagnosis functions prevent the device malfunction to ensure reliability.
- It detect the system error in memory, internal voltage level, communication and external power supply.

RS485/Modbus Communication

- It supports Modbus/RTU, major industrial communication protocol by RS485 Multi-drop to check the measured value and monitor the system remotely.



Bluetooth Communication

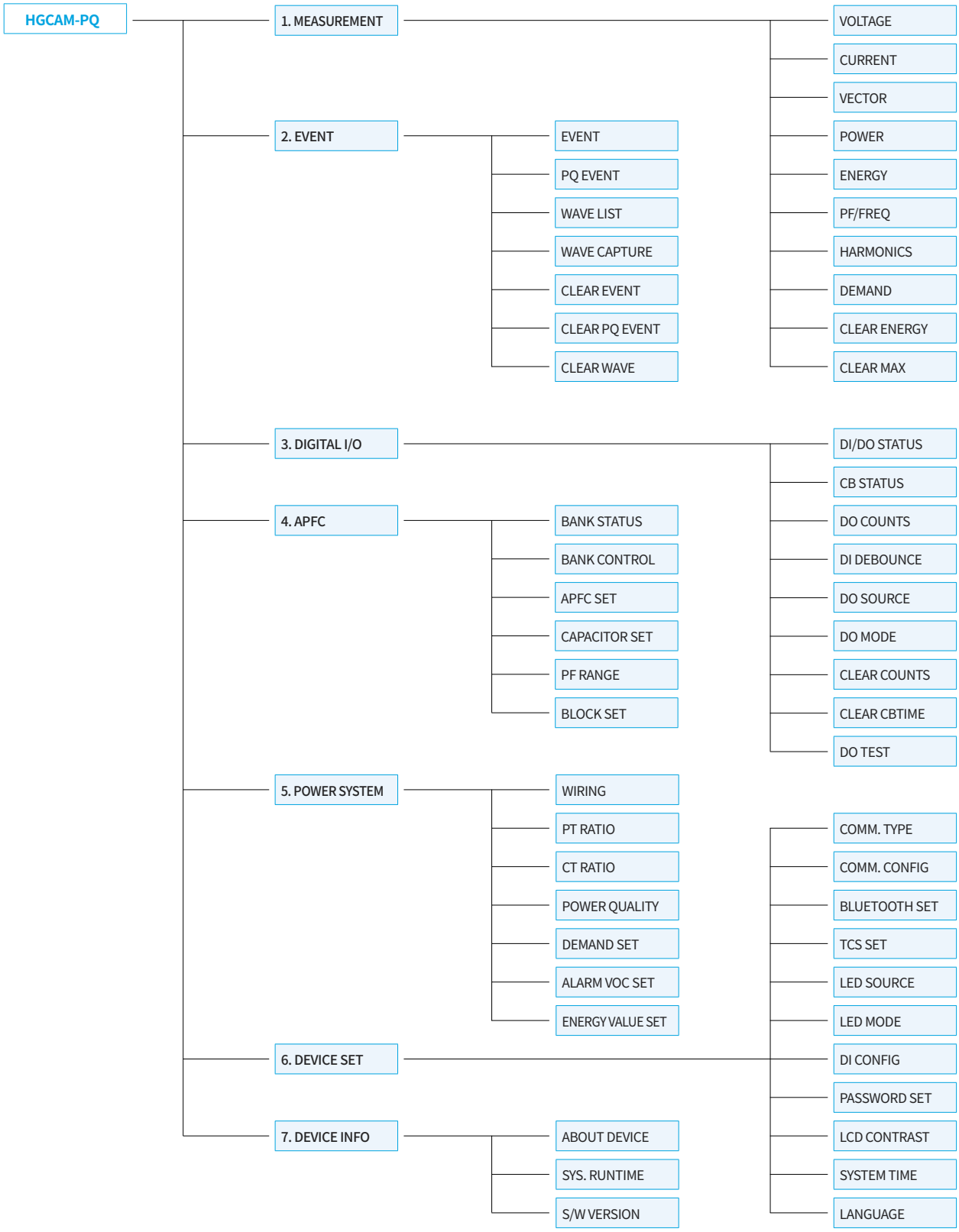
- User can check the measured data and event records safely and easily using smartphone manage application by connecting your personal smartphone through Bluetooth communication without a separate PC or communication equipment.
- Search "HG CB Link" on Google Play™ and download it for free on Android 7.0 and later smartphone.



Technical Data

Control and Setting

HGCAM-PQ



General Control

1. Enter Setting Menu

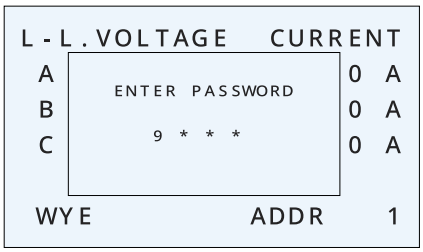
On the measurement screen, pressing the ESC button changes the setting menu. Press the ESC button again to return to the measurement screen.



2. Enter Password

The setting lock function runs by default to prevent unauthorized operation, so it is impossible to change the setting value or operate the circuit breaker in the locked state.
To make the setting change, you must enter the correct password to unlock it.
If you are performing an operation that requires unlocking while locked, a pop-up window will appear asking for your password.

< > buttons : move the cursor's position.
V ∧ buttons : increase/decrease the value of each digit.
Enter 4 digits of the password and press the ENTER button.
The unlocked state is valid for 2 minutes.



The default password is 9999. For safe operation of the switchboard, please change it to your own password.
The password can be changed in the 6. DEVICE SET ▷ 8. PASSWORD SET in the Settings menu.

Technical Data

Control and Setting

General Control

3. System Setting

The system settings are composed of the POWER SYSTEM setting for the switchboard environment and the DEVICE SET setting for the meter's own operating conditions.

POWER SYSTEM

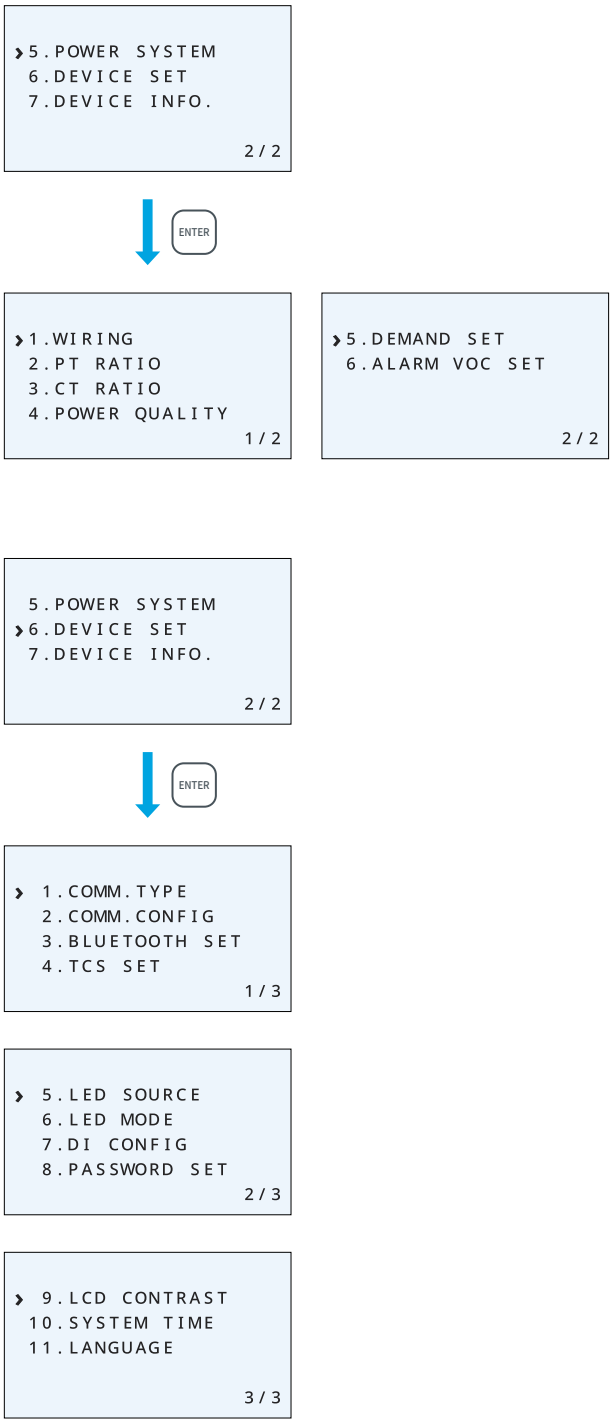
No.	Menu	Description	Value
1	WIRING	PT Secondary Wiring Type	WYE, DELTA
		PT Secondary Wiring FACTOR	1/√3, 1, ADV
2	PT RATIO	PT Primary Rating	110 ~ 345,000V (10step)
		PT Secondary Rating	100 ~ 440V (10step)
3	CT RATIO	CT/NCT Primary Rating	5 ~ 9,000A (1step)
		CT/NCT Secondary Rating	5A(Fixed)
4	POWER QUALITY	Sag Use Set	ON, OFF
		Sag Pickup Set	0.40 ~ 0.90Vn (0.01step)
		Swell Use Set	ON, OFF
		Swell Pickup Set	1.10 ~ 2.00Vn (0.01step)
		Interruption Use Set	ON, OFF
		Interruption Pickup Set	0.10 ~ 0.30Vn (0.01step)
5	DEMAND SET	Interval	1 ~ 60min (1step)
		Update Type	Sliding Block, Timed Block
		Clear Period	None, Every Day, Every Month
		Demand Current	0.10 ~ 1.20In (0.01step)
		Demand Active Power	0.10 ~ 1.20Pn (0.01step)
		Demand Reative Power	0.10 ~ 1.20Qn (0.01step)
		Trend Log Set	ON, OFF
6	ALARM VOC SET	Alarm VOC stage	#1~#4 setting
		Channel	Va, Vb, Vc, Vab, Vbc, Vca, V1, V2, V0, Ia, Ib, Ic, Io, I1, I2, I0, P, Q, S, Hz, PF
		Limit Type	Lower, Upper
		Delay Time	0.10 ~ 60.00sec (0.01step)
		Pickup Level	0 ~ 99,999,999 (1step) Hz, PF : 0~ 999,999.99 (0.01step)
		Dead Band Level	0 ~ 99,999,999 (1step) Hz, PF : 0 ~ 999,999.99 (0.01step)
7	ENERGY VALUE SET	Es, Esa, Esb, Esc	0 ~ 99,999,999kVAh (1step)
		+Ep, +Epa, +Epb, +Epc	0 ~ 99,999,999kWh (1step)
		-Ep, -Epa, -Epb, -Epc	0 ~ 99,999,999kWh (1step)
		+Eq, +Eqa, +Eqb, +Eqc	0 ~ 99,999,999kVarh (1step)
		-Eq, -Eqa, -Eqb, -Eqc	0 ~ 99,999,999kVarh (1step)

※ PT primary rating is based on line voltage, and secondary rating is based on phase voltage.

DEVICE SET

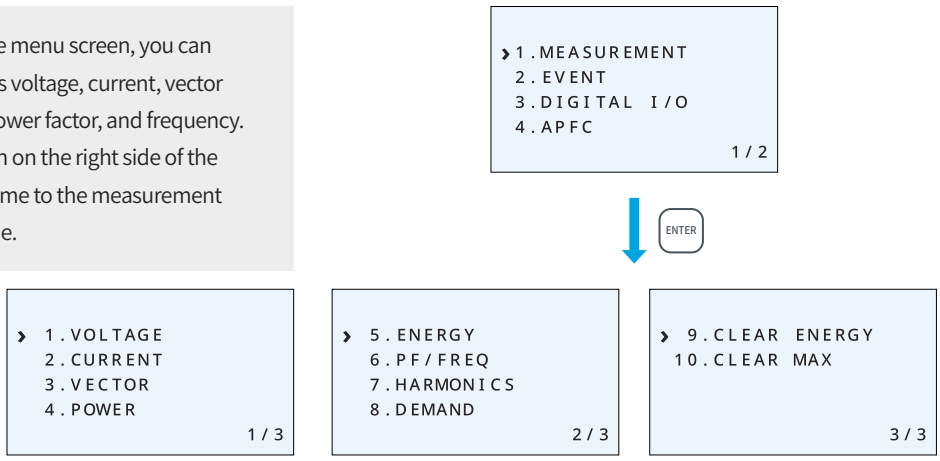
No.	Menu	Description	Value
1	COMM. TYPE	Communication Type Set	MODBUS-RTU
2	COMM. CONFIG	ADDR : Address	1 ~ 247
		BAUD : Baudrate	9600, 19200, 38400, 57600
		SWAP : SWAP WORD activate (Float) 1)	ON, OFF
3	BLUETOOTH SET	Bluetooth Use Set	ON, OFF
4	TCS SET	TCS Use Set	ON, OFF
		TCS Time Set	0.10 ~ 20.00sec (0.01step)
5	LED SOURCE	User-define LED01~08 source	System State, Alarm, DI State, etc.
6	LED MODE	IOperating Mode LED01~08	NORMAL, LATCH
7	DI CONFIG	DI01 Remote Set	SIGNAL, REMOTE
8	PASSWORD SET	User Password Set	-
9	LCD CONTRAST	LCD Display Contrast Set	-
10	SYTEM TIME	System Time(RTC) Set	
11	LANGUAGE	System Language Set	ENGLISH, KOREAN

※ 1) Endian swap is only for float data type. It works as SWAP WORD
ex) 1234.57 : 3D 52 9A 44 (SWAP OFF) → 9A 44 3D 52 (SWAP ON)



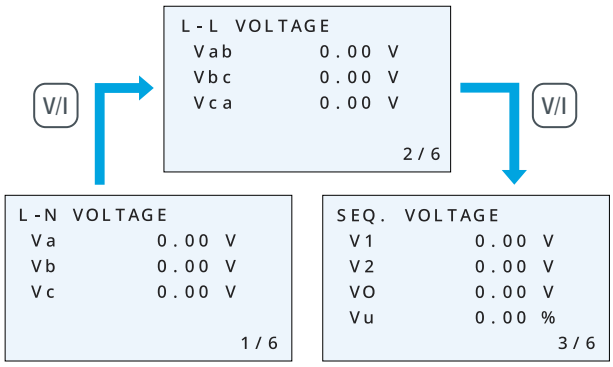
4. Measurement

In the 1. MEASUREMENT on the menu screen, you can check measured values such as voltage, current, vector diagram, power and energy, power factor, and frequency. By pressing the shortcut button on the right side of the LCD display screen, you can come to the measurement confirmation screen at any time.



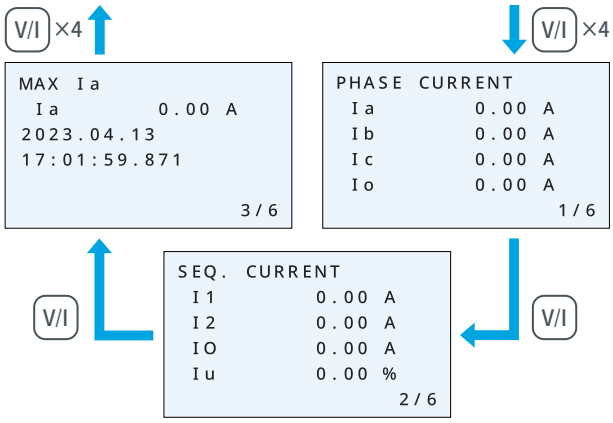
① VOLTAGE

- It displays phase voltage, line-to-line voltage, positive/negative/zero sequence voltage, and maximum voltage value for each phase.
- Press the shortcut button V/I to repeat the voltage and current items in order with each iteration.



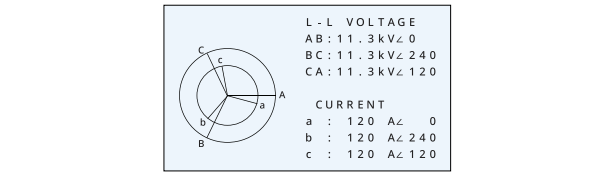
② CURRENT

- It displays phase current, positive/negative/zero sequence current, and maximum current value for each phase.
- Press the shortcut button V/I to repeat the voltage and current items in order with each iteration.



③ VECTOR

- It displays the magitude and angle of phase voltage and phase current as form of vector diagram.
- It can also be moved to the screen directly by pressing the shortcut button .



Technical Data

Control and Setting

General Control

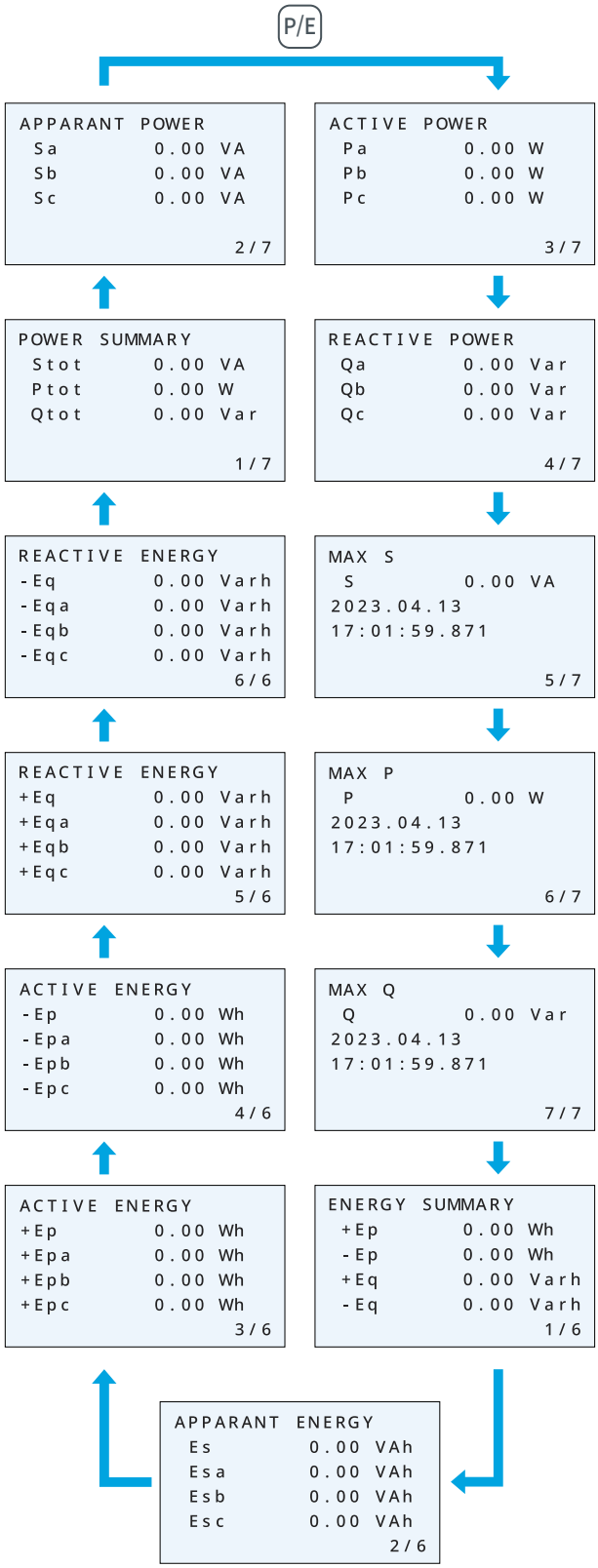
4. Measurement

④ **POWER**

- It displays active/reactive power, power factor and frequency.
- It can also be moved to the screen directly by pressing the shortcut button P/E.

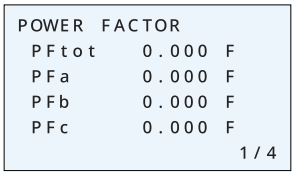
⑤ **ENERGY**

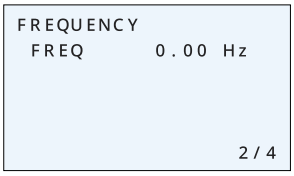
- It displays active/reactive energy, reverse active/reactive energy and apparent energy.
- It can also be moved to the screen directly by pressing the shortcut button P/E.

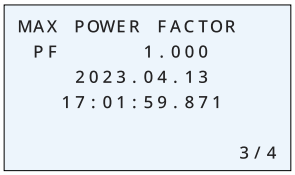


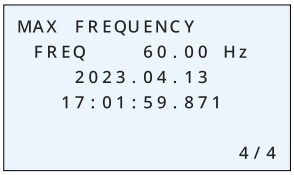
⑥ PF / FREQ

- It displays power factor and frequency.



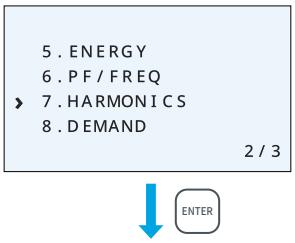


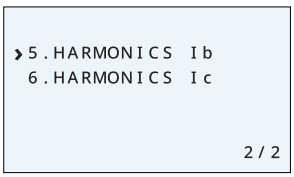




⑦ HARMONICS

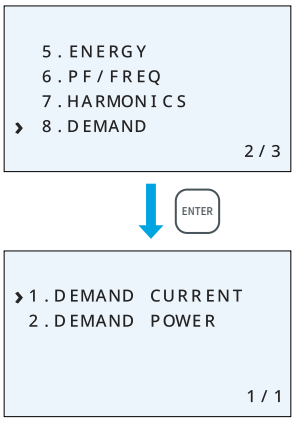
- It displays harmonics of voltage and current up to 15th.
- THD, TDD, K-factor can be checked.





⑧ DEMAND

- It displays demand current, demand active power and demand reactive power.



⑨ CLEAR ENERGY

- The cumulative energy value can be cleared.

⑩ CLEAR MAX

- The maximum value of voltage, current, power, power factor, frequency and demand can be cleared individually.

Technical Data

Control and Setting

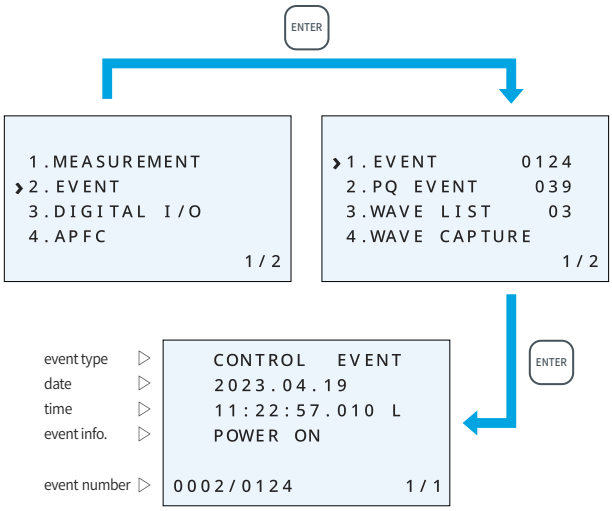
General Control

5. Event Log

In 2.EVENT menu, detailed general event and PQ event can be checked.

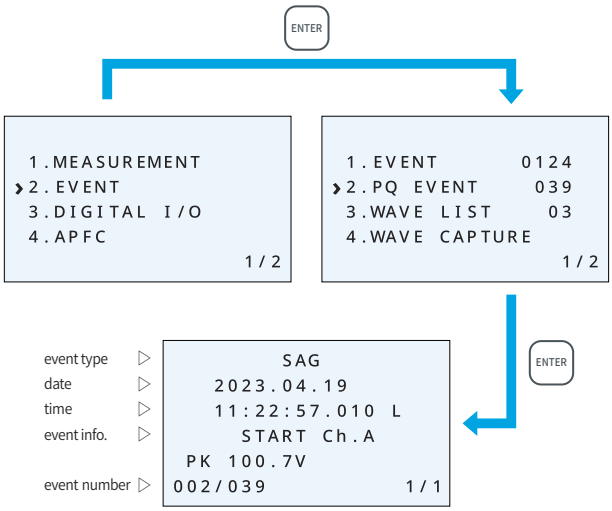
General Event

- User can check the general events occurring for various status and setting changes of the product in menu **2.EVENT → 1.EVENT**.
- Details of the event and the time when the event occurs can be inquired.
- Up to 1024 event records are stored, and each additional event is deleted from the oldest event history.
- **5.CLEAR EVENT** menu allow you to initialize the event history.



PQ Event

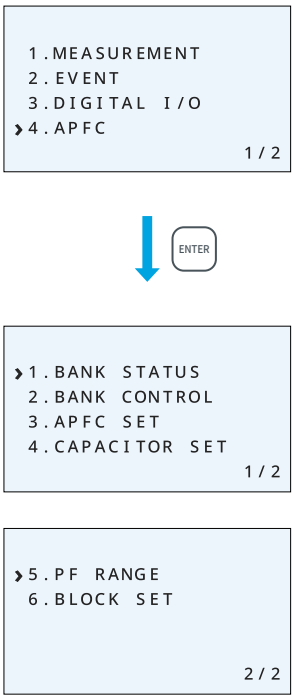
- User can check the PQ events for power quality management in menu **2.EVENT → 2.PQ EVENT**.
- Details of PQ element, voltage, time information is useful for system power factor management.
- Up to 256 PQ event records are stored, and each additional event is deleted from the oldest PQ event history.
- **6.CLEAR PQ EVENT** menu allow you to initialize the PQ event history.



6. Automatic Power Factor Control(APFC)

Detailed operating condition of APFC is set in 4.APFC menu.

No.	Menu	Description	Value
1	BANK STATUS	P	Real-time active power
		Q	Real-time reactive power
		PF	Real-time power factor
		BANK STATUS	BANK state (ON : ○, OFF : |)
2	BANK CONTROL	BANK 01 Control	ON, OFF
		BANK 02 Control	ON, OFF
		BANK 03 Control	ON, OFF
		BANK 04 Control	ON, OFF
		BANK 05 Control	ON, OFF
		BANK 06 Control	ON, OFF
3	APFC SET	APFC Use Set	ON, OFF
		Mode Set	AUTO, MANUAL, LOOP
		Number of Condensor	1 ~ 6 (1step)
		Delay Time Set	3 ~ 3,000sec (1step)
		Dead Time Set	3 ~ 3,000sec (1step)
4	CAPACITOR SET	BANK 01 Capacity Set	0 ~ 99,999,999VA (1step)
		BANK 02 Capacity Set	0 ~ 99,999,999VA (1step)
		BANK 03 Capacity Set	0 ~ 99,999,999VA (1step)
		BANK 04 Capacity Set	0 ~ 99,999,999VA (1step)
		BANK 05 Capacity Set	0 ~ 99,999,999VA (1step)
		BANK 06 Capacity Set	0 ~ 99,999,999VA (1step)
5	PF RANGE	Max. PF Set	0.90 ~ -0.95 (0.01step)
		Min. PF Set	0.70 ~ 0.95 (0.01step)
		PF Alarm Set	0.00 ~ 0.90 (0.01step)
6	BLOCK SET	Under Current BLOCK Set	0.10 ~ 1.00In (0.01step)
		Under Voltage BLOCK Set	0.50 ~ 1.00Vn (0.01step)
		Over Voltage THD BLOCK Set	3 ~ 50% (1step)



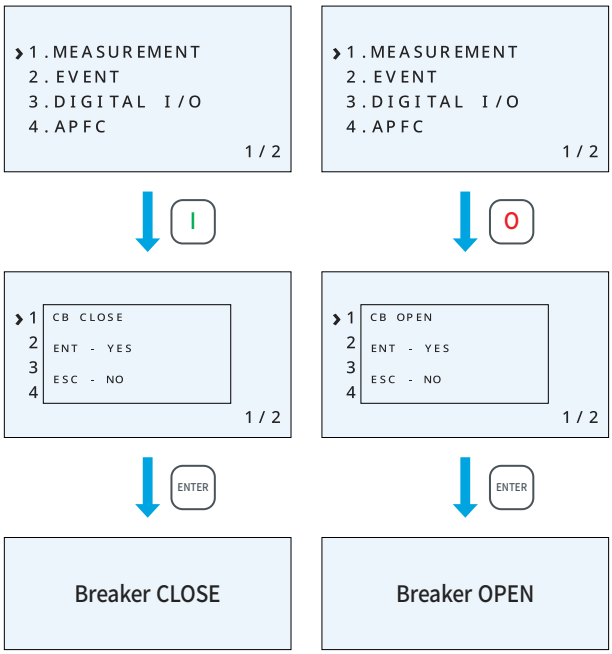
Technical Data

Control and Setting

General Control

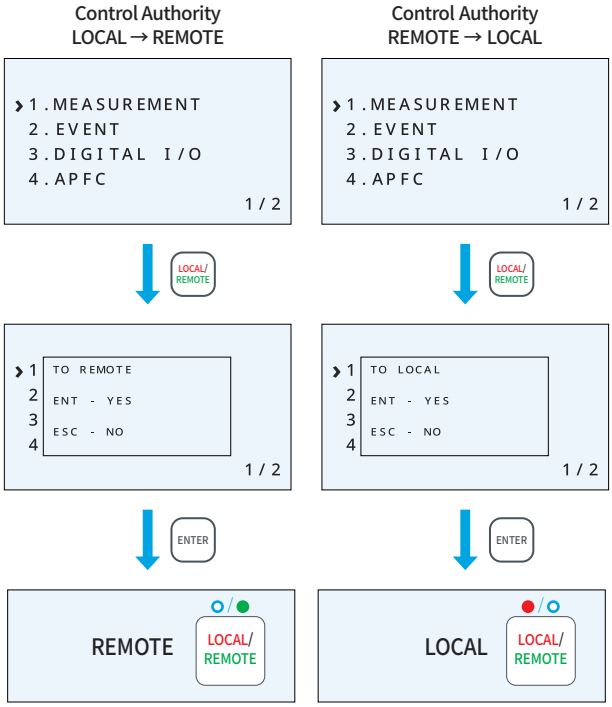
7. Breaker Control

- On any screen, user can directly control the opening and closing of the circuit breaker.
- For the breaker control, the control authority must be in LOCAL state, and it can be controlled after the password is released first. (Check the LED on LOCAL/REMOTE button is red)



8. Control Authority Set

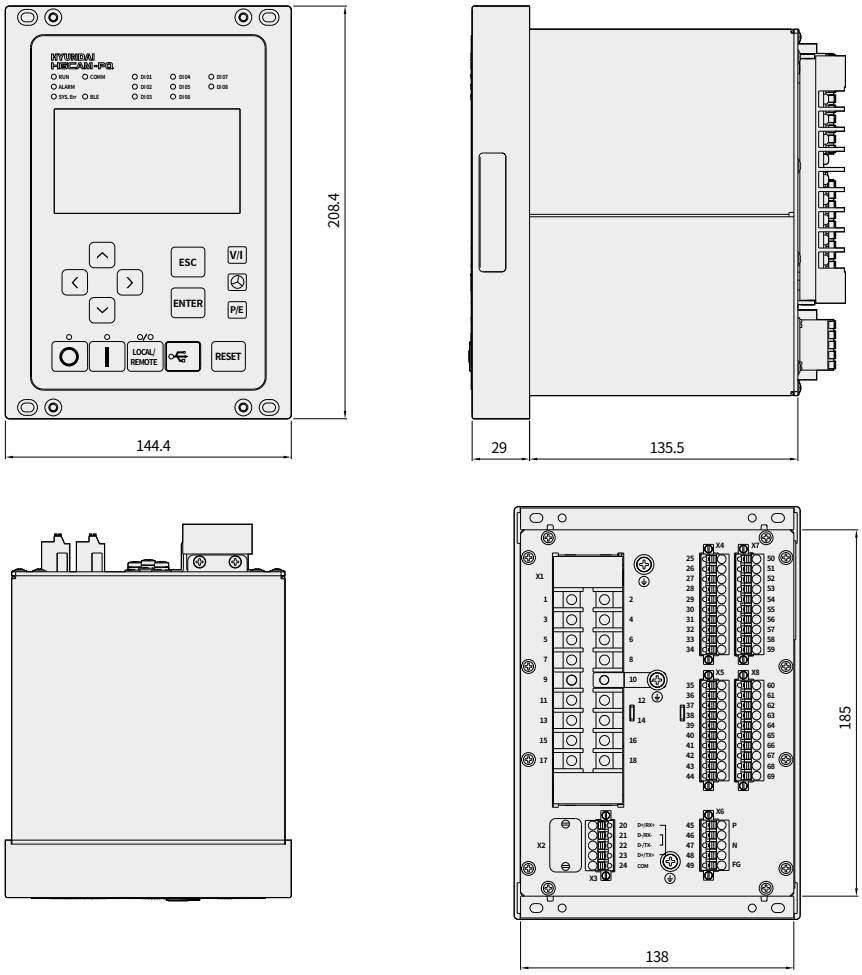
- On any screen, user can switch control authority to LOCAL↔REMOTE using LOCAL/REMOTE button. In order to control the relay remotely with the communication, the control authoriry must be switched to REMOTE.
- In the REMOTE state, setting changes and relay control using the physical button are disabled.
- If the control authority of the relay is LOCAL, the red LED at top of the button is lit, and if it is REMOTE, the green LED is lit.



Dimension

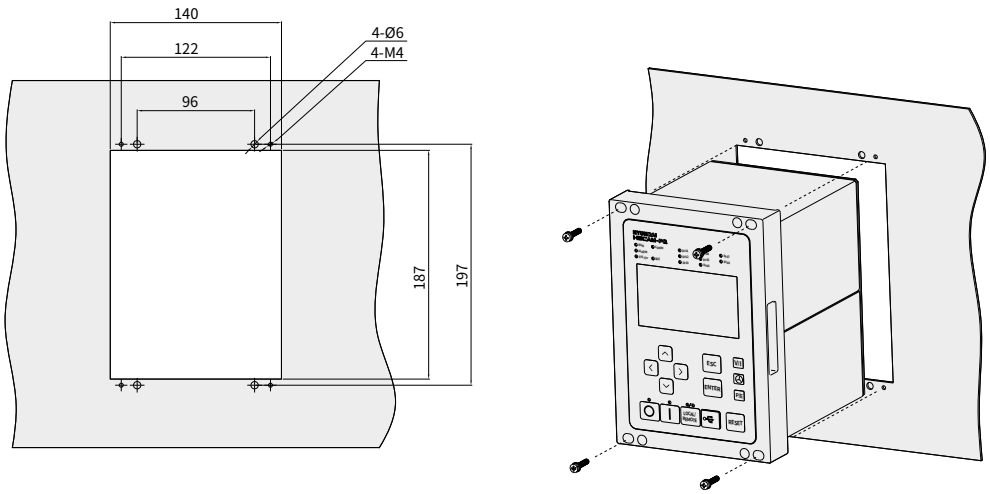
Outline Dimension

unit: mm



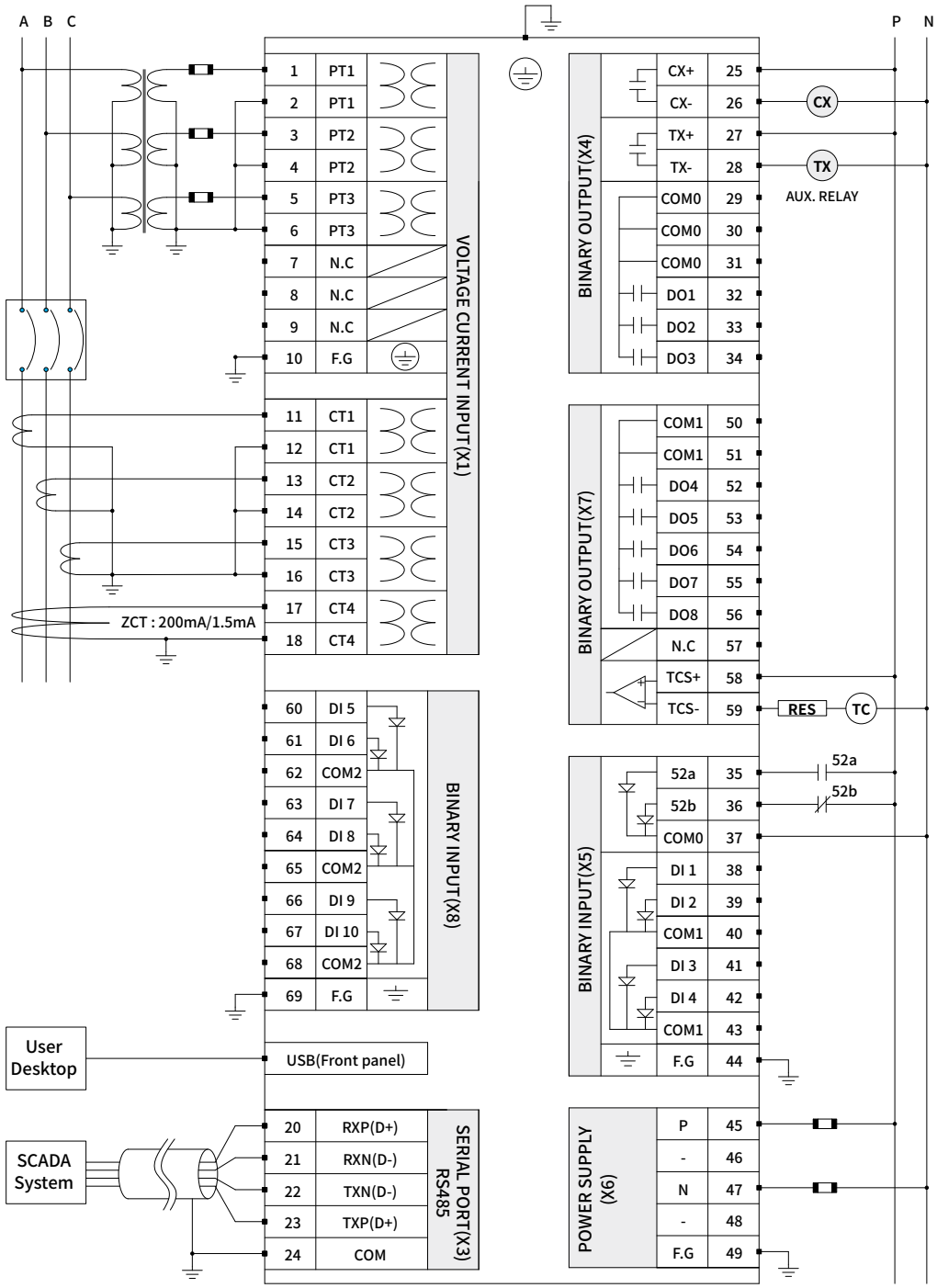
Panel Cutting Size

unit: mm



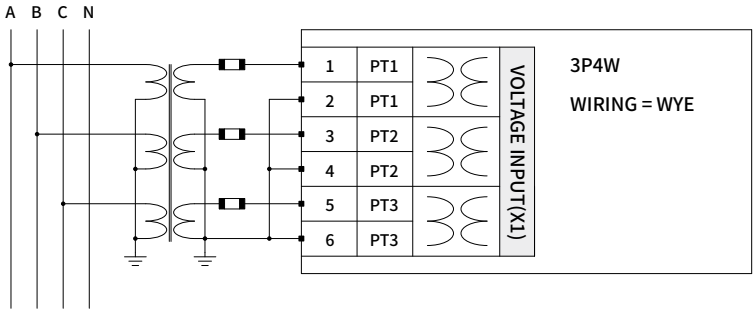
Wiring Diagram

General Wiring Diagram

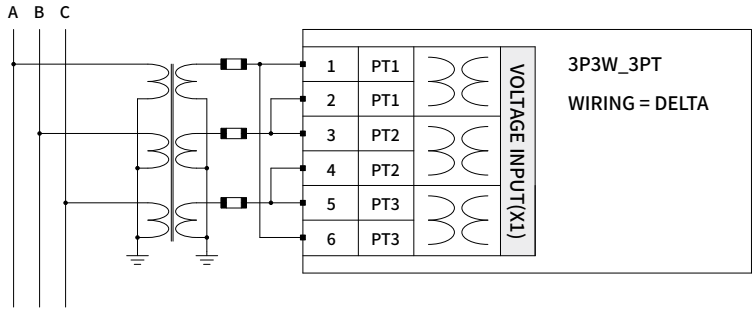


Wiring Diagram(PT)

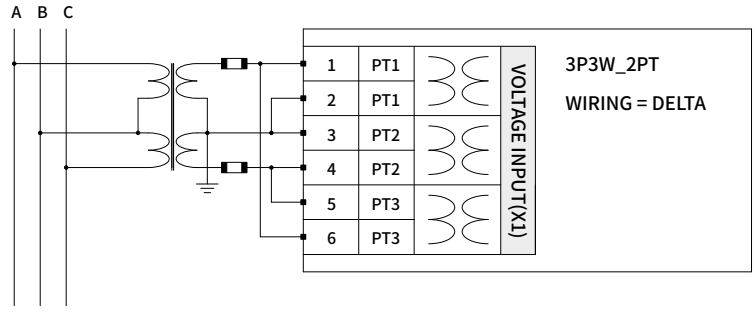
3P4W



3P3W 3PT

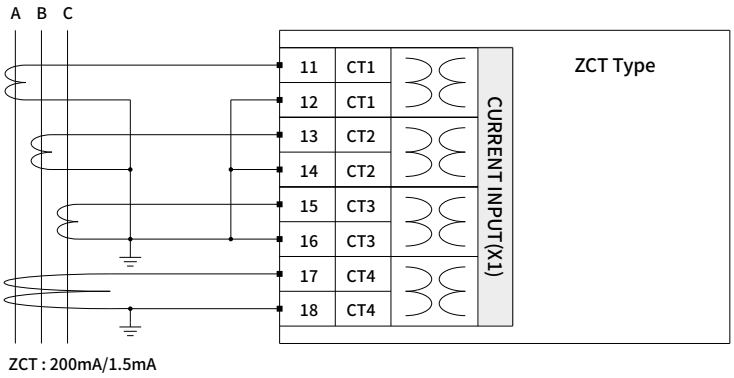


3P3W 2PT

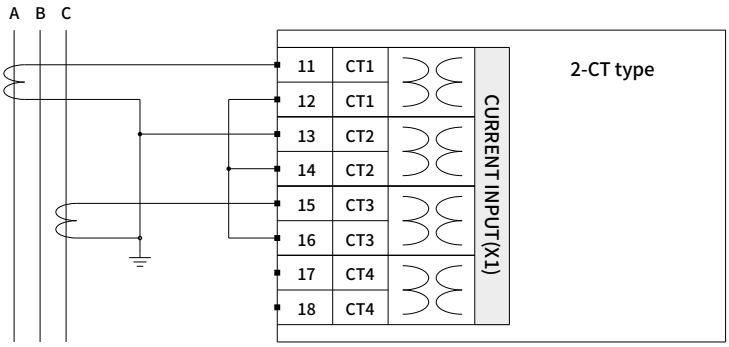


Wiring Diagram(CT)

3P3W ZCT

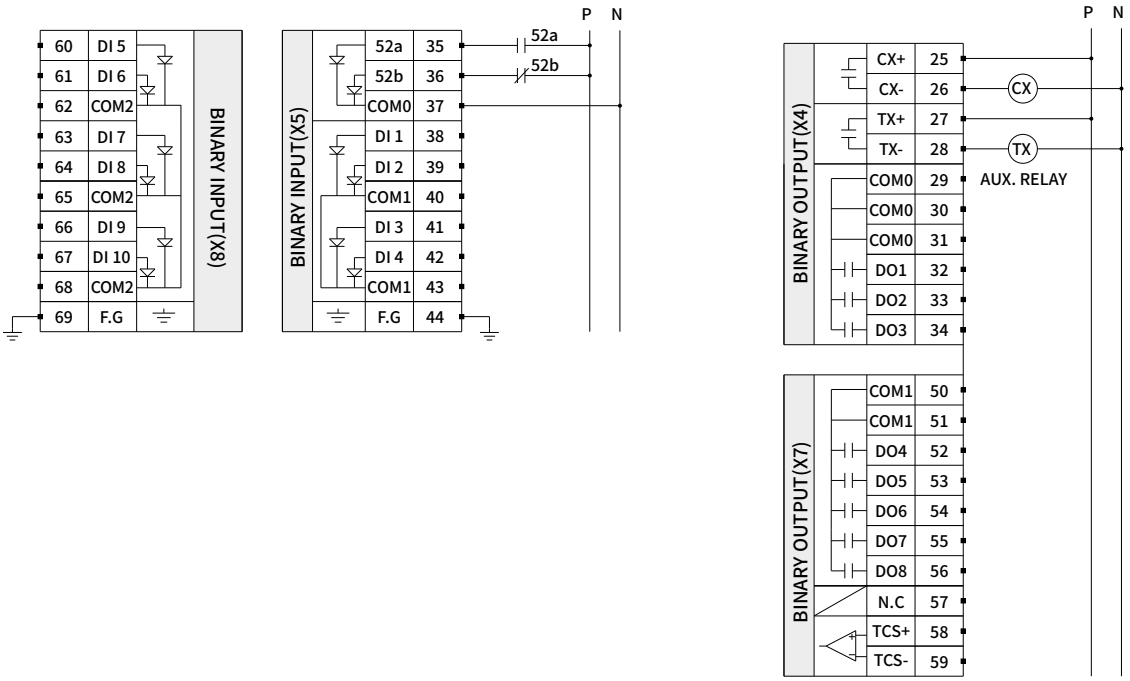


2CT

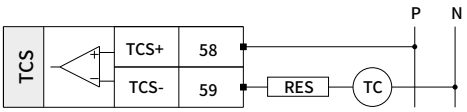


Wiring Diagram(DI/DO)

DI/DO



TCS



※ The RES(resistor) value is recommended to be 15 times of coil DC resistance. (over 10 Watt)
※ TC : Trip coil of circuit breaker (52b)

Order Code

HGCAM-PQ

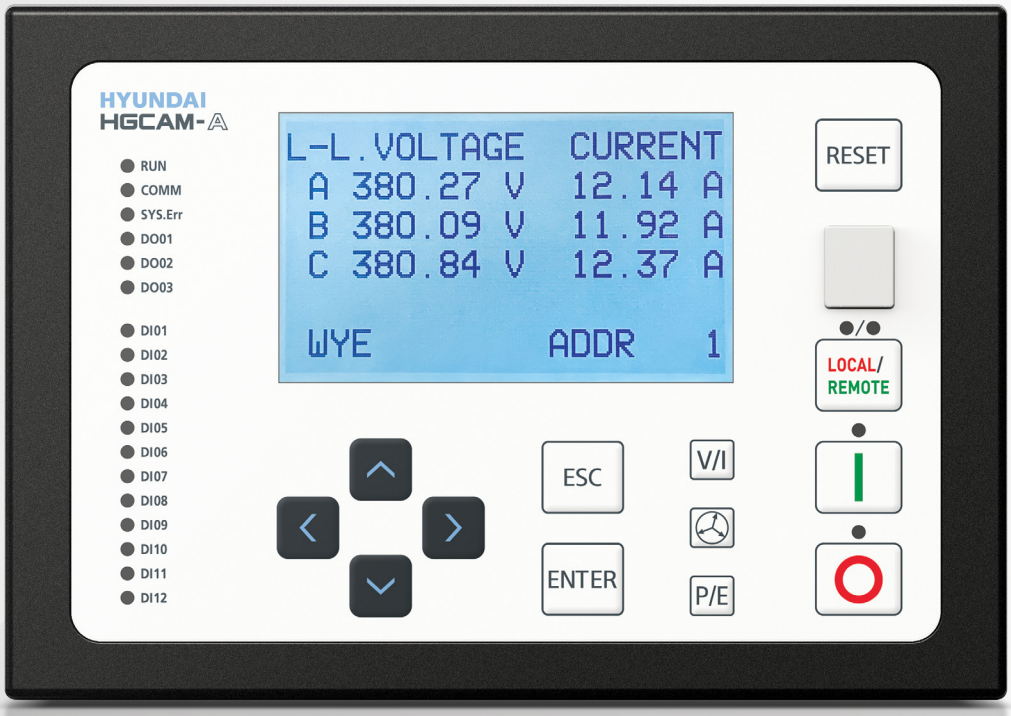
HGCAM	P	N	A5	F6	C1	NO	NO
Category	Type	Outfit	CT 2nd Rating	Frequency	Comm.	I/O module	Optional
HGCAM Digital Power Meter	**P** Power Quality	**N** Rectangular	**A5** AC 5A	**F6** 60Hz	**C1** MODBUS/ RS485	**NO** No Option	**NO** No Option

HG Series

Digital Power Meter

HGCAM-A

Viable Choice for Power Measurement of Switchgear



Overview

Measurement Accuracy

HGCAM-A power meter comply with IEC62053, a high-precision measurement standard. For rated voltage of 10V to 440V and rated current of 0.05A to 6A, it meets F/S accuracy 0.3% and for power and energy, it meets F/S accuracy 0.5%.

Enhanced Measurement

It measures, analyzes and displays power quality such as three-phase voltage, current, power and energy, power factor, frequency, demand, harmonics up to 15th, THD, and TDD of the power system.

Real-time Communication

It supports Modbus/RTU, major industrial communication protocol by RS485 Multi-drop to check the measured value and monitor the system remotely.

Alarm Features

It monitor the power system abnormalities continuously such as erroneous lines, frequency anomalies, PT Fuse Fail, and CB operation control Fail.

Self-diagnosis

The self-diagnosis function continuously monitors the condition of the device and power system. And when the error occurs, it is notified to user.

Breaker Control

It is possible to control the connected circuit breaker by the circuit breaker ON/OFF contact, and it supports local control through physical buttons of front side and remote control using Modbus communication. Also SBO/CBO function support safe breaker control.

Convenient Replacement

It is easy to replace the product by applying the same dimension as discontinued product.

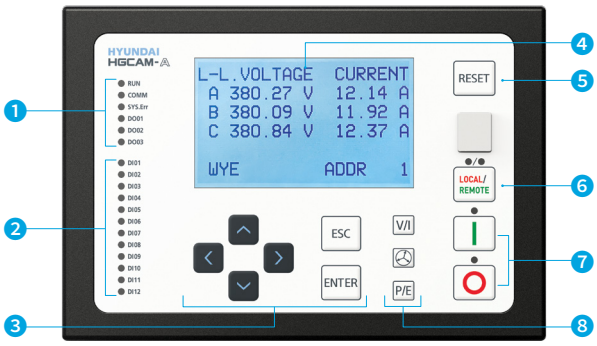


Appearance

Digital Power Meter (HGCAM-A)

Front View

No.	Classification	Description
1	Status LED	Run(●) Communication(●) System Error(●) DO contact(●)
2	DI Status LED	Alarm for each DI operation(●)
3	Function Key	Move between pages change setting value Enter menu page confirm save
4	LCD Display	Display measurement setting and record
5	RESET Key	Clear the alarm
6	LOCAL/REMOTE	Local/remote control to toggle permission Local(●)/Remote(●) state LED
7	CB ON/OFF	CB Close(●)/Open(●) state LED CB Close(❙)/Open(○) Key
8	Measure Hotkey	Shortcuts for each measurement (voltage/current, vector diagram, power/energy)

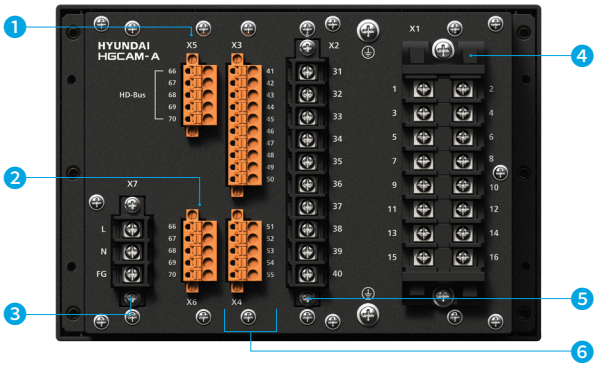


Key	Menu	Function
Arrow Key (Up, Down)	Display Menu	Move between item
	Setting Menu	Move between item, Change setting value
Arrow Key (Left, Right)	Display Menu	Move to submenu
	Setting Menu	Move between item, Change setting value
MENU Key	Display Menu	Move to Setting Menu
	Setting Menu	Move to Display Menu
RESET Key	Sys.Err mode	Return to normal display
	Alarm mode	DO state reset
ESC Key	Setting Menu	Move to top menu
	Save mode	Cancel the changed data
ENTER Key	Password mode	Enter the password
	Save mode	Save the changed data
LOCAL/REMOTE Key	All	Toggle local/remote control permission
CLOSE/OPEN Key	All	Manual control for circuit breaker

Terminal Block Configuration

Rear View

No.	Classification	Description
1	HDBus Comm.	RS-485, HDLC, 62.5, 125kbps
2	ModBus Comm.	RS-485 9600, 19200, 38400, 57600bps
3	Control Power	AC/DC 90 ~ 265V Free Voltage
4	CT, PT	CT : 4 PT : 4
5	DO	Communication : 2 CB Control : 2
6	DI	CB Monitoring : 2 Remote : 1, General : 9



X1 (Voltage/Current)	
1, 2	CT1 input
3, 4	CT2 input
5, 6	CT3 input
7, 8	CT4 input
9, 10	PT1 input
11, 12	PT2 input
13, 14	PT3 input
15, 16	PT4 input

X2 (DO)	
31, 32	DO01
33, 34	DO02
35, 36	DO03
37, 38	DO04 (CB on)
39, 40	DO05 (OB off)

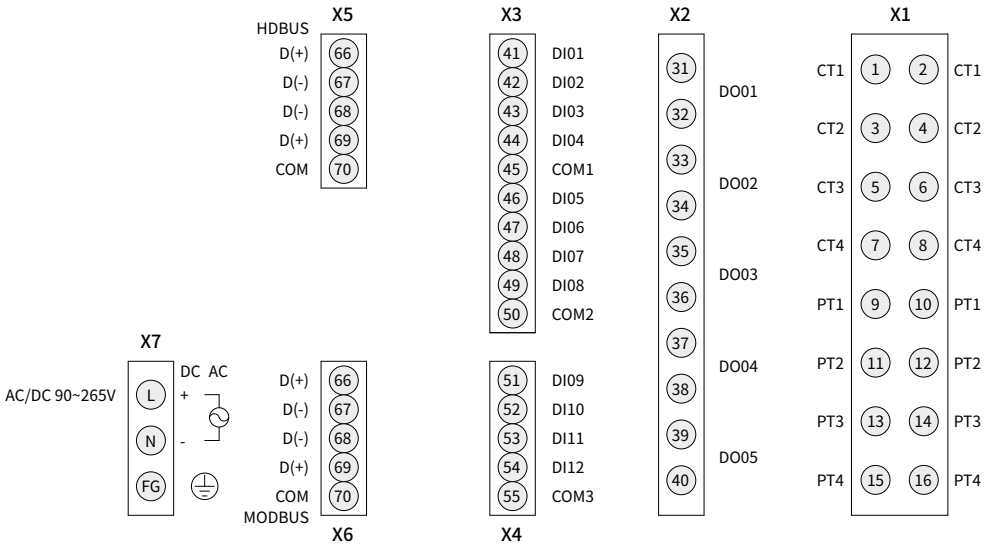
X3 (DI)	
41, 45	DI01 (CB ON)
42, 45	DI02 (CB OFF)
43, 45	DI03 (Remote)
44, 45	DI04
46, 50	DI05
47, 50	DI06
48, 50	DI07
49, 50	DI08

X4 (DI)	
51, 55	DI09
52, 55	DI10
53, 55	DI11
54, 55	DI12

X6 (Comm.)		
66, 69	D+	MODBUS Comm.
67, 68	D-	
70	COM	

X5 (Comm.)		
66, 69	D+	HDBUS Comm.
67, 68	D-	
70	COM	

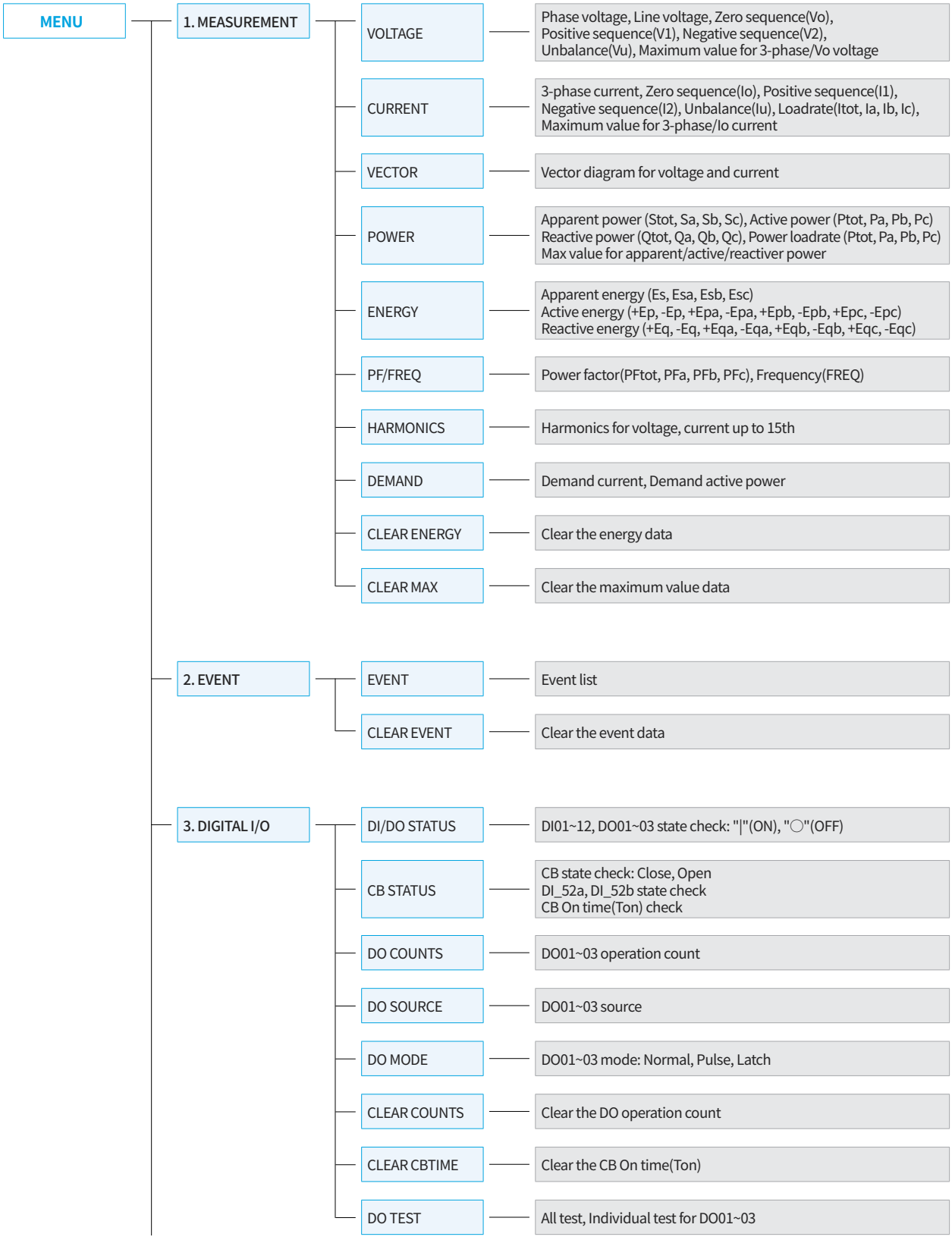
X7 (Control Power)	
L, N	Control power
FG	Ground



Technical Data

Control and Setting

Menu Tree

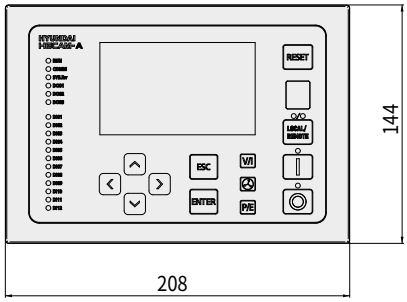


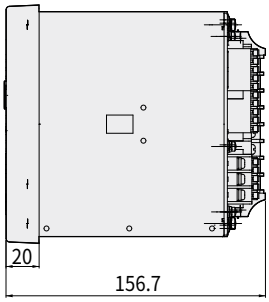
4. POWER SYSTEM	WIRING	PTPT 2nd wiring: Wye, Delta Factor set: 1/√3, 1, ADV
	PT RATIO	PT/GPT 1st rating: 100~345000V (line voltage) PT/GPT 2nd rating: 100~440V (line voltage)
	CT RATIO	CT/NCT 1st rating: 5~9000A CT/NCT 2nd rating: 5A (fixed)
	DEMAND SET	Demand interval: 1~60min Update type: Sliding Block, Timed Bock Initial period: None, Every Day, Every Month Pickup current: 0.10~1.20 In Demand active power: 0.10~1.20 Pn Demand reactive power: 0.10~1.20 Qn
	ALARM VOC SET	Analog VOC stage set: #1~#4 Stage set: Va, Vb, Vc, Vn, Vab, Vbc, Vca, V1, V2, V0, Ia, Ib, Ic, Io, I1, I2, I0, P, Q, S, Hz, PF, LP, LI Limit type: Lower, Upper Delay time: 0.10 ~ 60.00 sec Pickup level: 0 ~ 99,999,999 (Hz, PF는 0.00 ~ 999,999.99) Dead band level: 0 ~ 99,999,999 (Hz, PF는 0.00 ~ 999,999.99)
	ENERGY VALUE SET	Es, Esa, Esb, Esc: 0 ~ 99,999,999 kVAh +Ep, +Epa, +Epb, +Epc: 0 ~ 99,999,999 kWh -Ep, -Epa, -Epb, -Epc: 0 ~ 99,999,999 kWh +Eq, +Eqa, +Eqb, +Eqc: 0 ~ 99,999,999 kVarh -Eq, -Eqa, -Eqb, -Eqc: 0 ~ 99,999,999 kVarh
5. DEVICE SET	COMM. TYPE	Comm. type set: MODBUS-RTU, HDBUS
	COMM. CONFIG	(COMM. TYPE = Modbus-RTU) ADDR: 1 ~ 247 BAUD: 9600, 19200, 38400, 57600 bps SWAP: Float data swap ON, OFF MAP: Modbus profile set HGCAM, HiCAMV1, HiCAMV2 (COMM. TYPE = HDBUS) ADDR: 1 ~ 247 BAUD: 62.5, 125 Kbps MAP: HDBUS profile set HGCAM, HiCAMV1, HiCAMV2
	DI CONFIG	DI03 Remote contact mode: SIGNAL, REMOTE
	PASSWORD SET	Set the password (Default set 9999)
	LCD CONTRAST	Contrast set for LCD display
	SYSTEM TIME	System time set
	LANGUAGE	Language set: English, Korean
6. DEVICE INFO	ABOUT DEVICE	Device name, Comm. type. CT 2nd rating, Frequency set
	SYS. RUNTIME	System accumulated runtime
	S/W VERSION	Firmware version

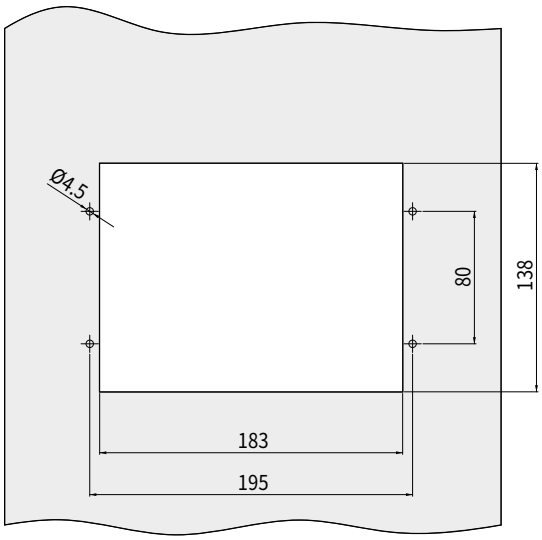
Dimension

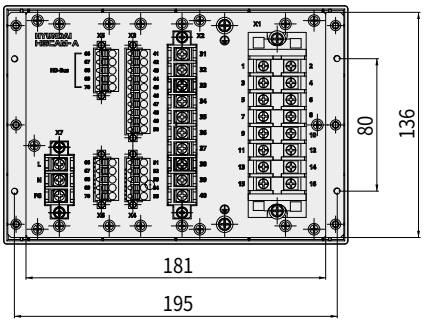
Outline and Panel Cutting

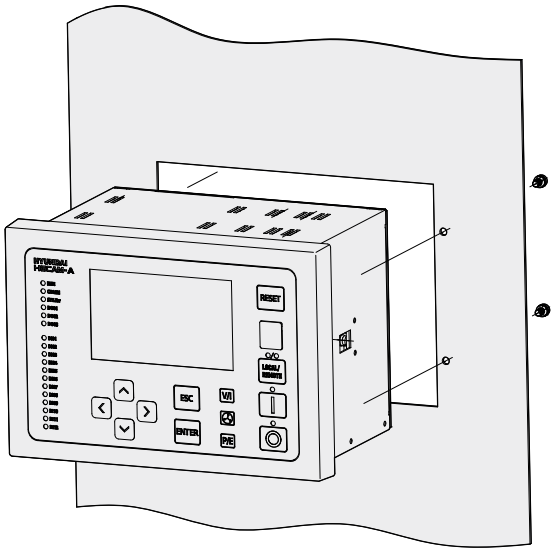
unit: mm

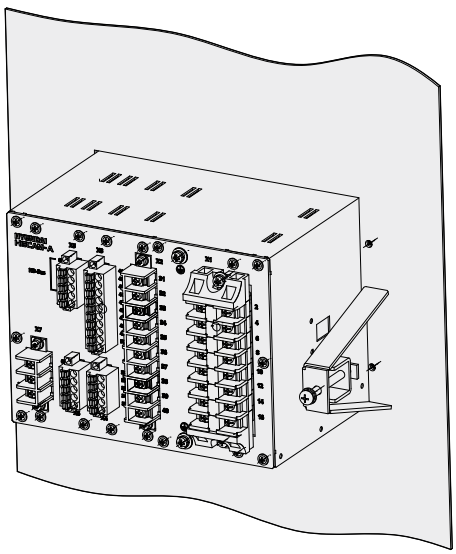






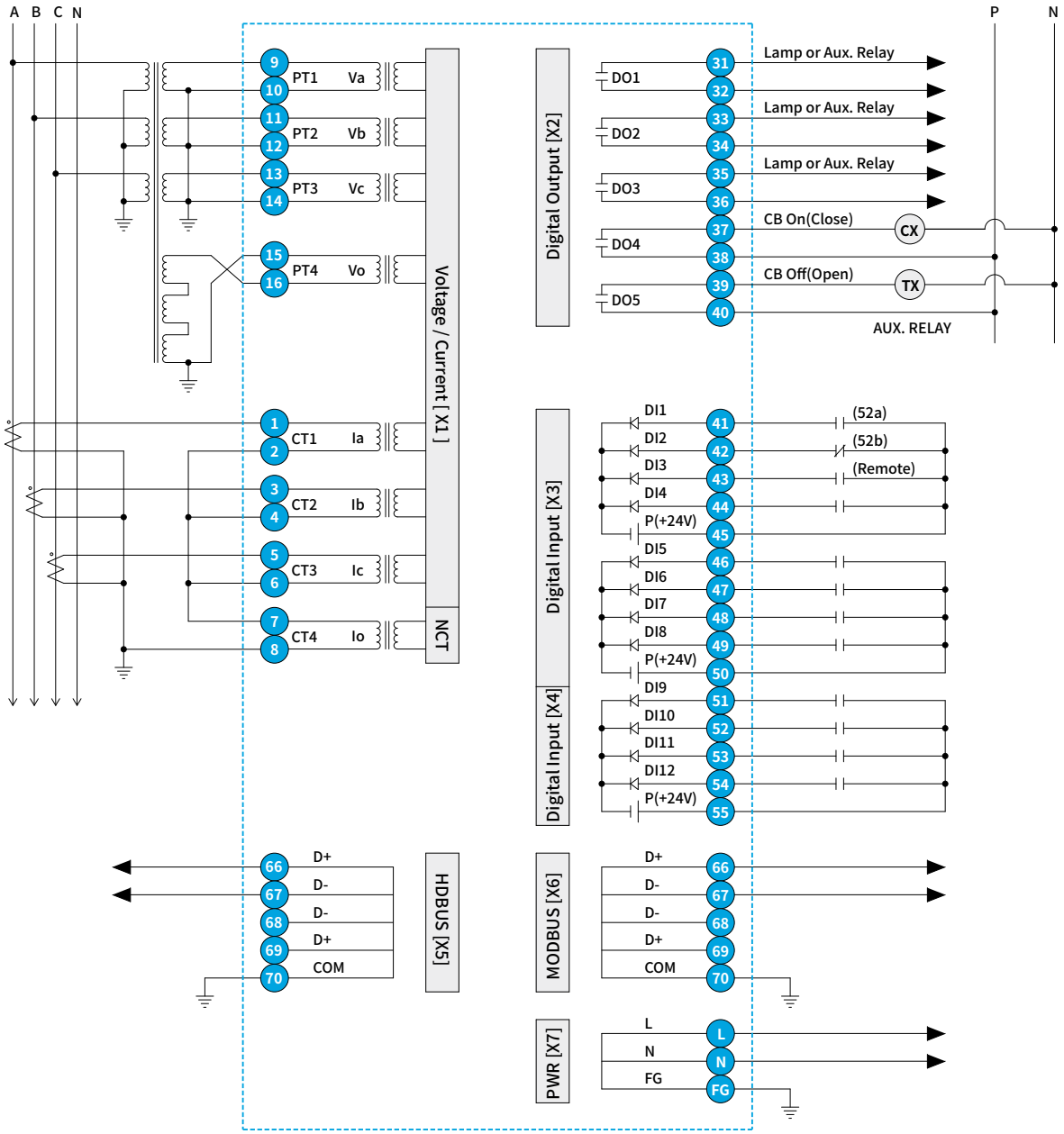






Wiring Diagram

General Wiring Diagram



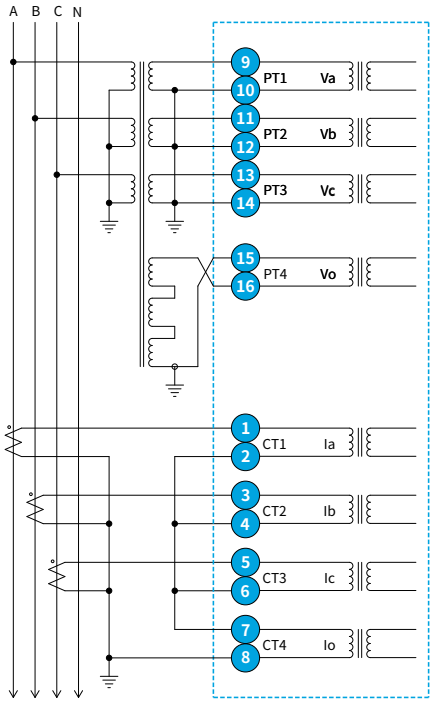
※ DO wiring
Be sure to use the auxiliary relay for circuit breaker

※ DI wiring
The default DI connection is described as a basis of DC 24V internal power. In the wiring diagram, X3:45, X3:50, and X4:55 terminals are common terminals of the DC 24V output, so it should not be connected to an external control power source.
It may cause Internal malfunction and device breakdown during DI connection with external control power.
Please find a dedicated option model to connect external control power. For more information, refer to DI wiring diagram on page 41.

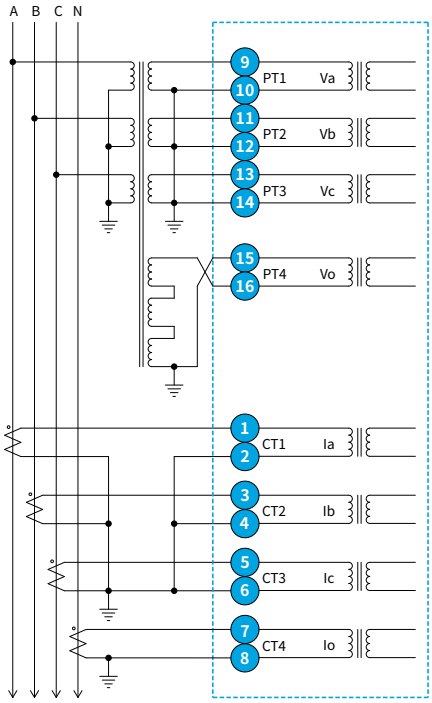
Wiring Diagram

Wiring Diagram(PT, CT)

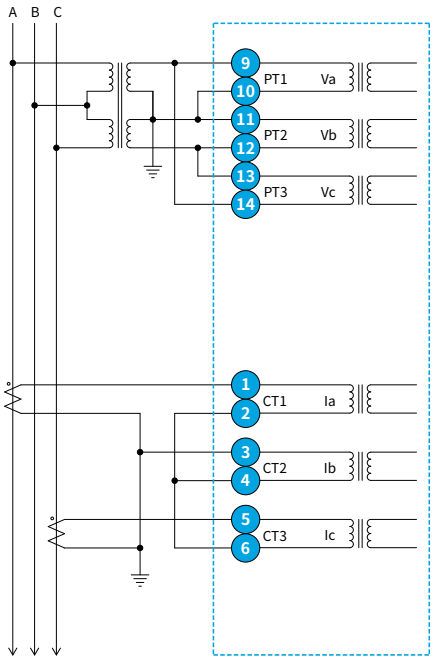
3P4W Residual Current



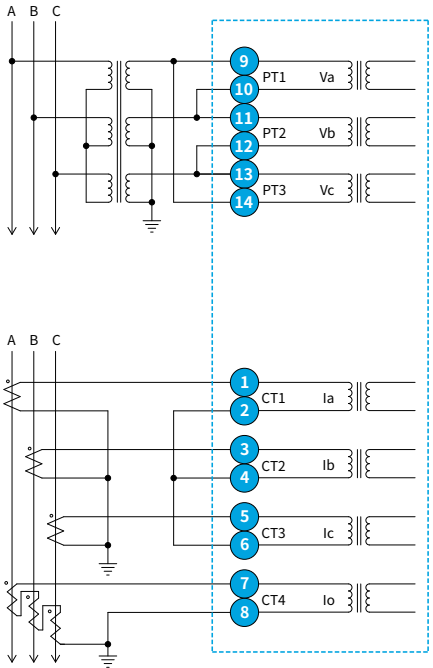
3P4W NCT



3P3W 2PT/2CT Delta

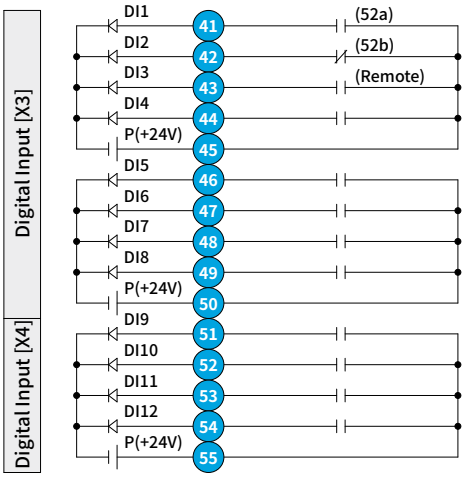


3PT Delta/3W CT

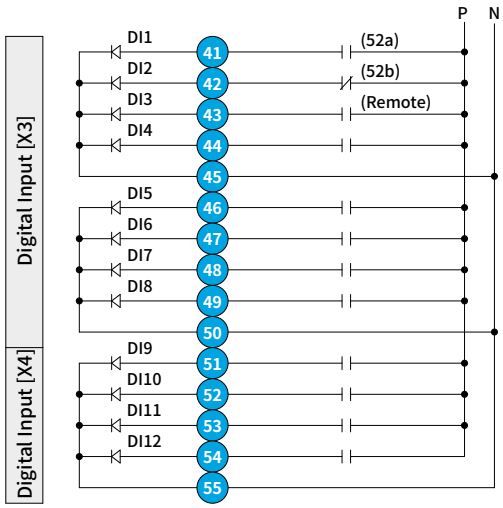


Wiring Diagram(DI)

24VDC internal power



90~265VAC/DC extenal power



※ Precautions for DI connection

· Only the DI connection method specified for each product is supported, and 24VDC internal power and 90~265VAC/DC external power cannot be mixed. Before connect the DI wiring, make sure to check that the type of model you purchased matches the drawing specifications.
- HGCAMANA5F6C5D2NO : 24VDC Internal Power Supply Type
- HGCAMANA5F6C5D3NO: 90-265VAC/DC External Power Supply Type

· External power using a CTD (Capacitor Trip Device) may cause malfunction of the relay, so please use the other power source.

Order Code

HGCAM-A

HGCAM	A	N	A5	F6	C5	D2	NO
Category	Type	Outfit	CT 2nd Rating	Frequency	Comm.	DI Type	Optional
HGCAM Digital Power Meter	**A** Advanced type	**N** Rectangular	**A5** AC 5A	**F6** 60Hz	**C5** Modbus-RTU, HDBUS	**D2** 24VDC Internal power **D3** 90 ~ 265V AC/DC External power	**NO** No Option

HG Series

Digital Power Meter

HGCAM-S

Compact Size & Strong Performance



Overview

Measurement Accuracy

HGCAM-S power meter comply with IEC62053, a high-precision measurement standard. For rated voltage of 10V to 440V and rated current of 0.05A to 6A, it meets accuracy 0.3% and for power and energy, it meets accuracy 0.5%.

Real-time Communication

It supports Modbus/RTU, major industrial communication protocol by RS485 Multi-drop to check the measured value and monitor the system remotely.

Various Measurement

It displays more than a dozen measurements, including 3-phase current, phase voltage and line voltage.

Wiring Diagnosis

It determine the direction of the voltage phase rotation and indicate to the user whether there is an abnormality in the connection to prevent reverse phase fault.

Convenient Installation

Compact design and easy wiring save installation space of panel.

Demand Control

Active power demand is set up with relay output functions and load current demand per 1kW.

Loadate display

Based on the set CT ratio and rated current, it displays a loadrate from 40% to 110% through LED lamp.

Remote ACB ON/OFF control

Using the MODBUS protocol, ACB breaker can be controlled remotely, and demand control feature is disabled during remote operation.



Appearance

Digital Power Meter (HGCAM-S)

Front View

No.	Classification	Description
1	FND Display	Display measurement and setting
2	Loadrate LED	Indicate the loadrate based on the rated current
3	Power, Power Factor	Indicate the apparent/active/reactive power and power factor
4	Comm., Phase Fail	Communication state and phase fail
5	Voltage Unit	Indicate the voltage unit
6	Current Unit	Indicate the current unit
7	Energy, Frequency	Indicate the active/reactive energy and frequency
8	Control Key	Parameter setting and display mode control



Rear View

Power, PT terminal	
FG	FG
L(+)	P
N(-)	N
Va	PT_Va
Vb	PT_Vb
Vc	PT_Vc
Vn	PT_Vn

CT terminal	
Ia1	Ia_IN
Ia2	Ia_OUT
Ib1	Ib_IN
Ib2	Ib_OUT
Ic1	Ic_IN
Ic2	Ic_OUT

Comm. terminal	
+	RS 485+
-	RS 485-

DI, DO terminal	
DI1	DI1
DI2	DI2
DI3	DI3
DI_COM	Common(P+)
DO1	DO1
COM1	COM1
DO2	DO2
COM2	COM2

※ Please connect the DI_COM terminal to a power source such as P+.
※ When using COM1 and COM2 terminals as CB ON/OFF, please connect power such as P+.



Basic Type

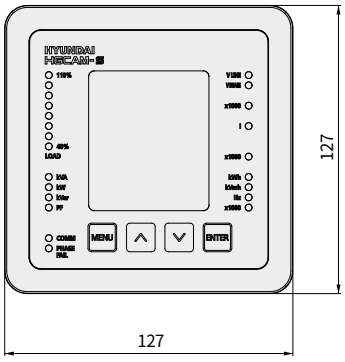


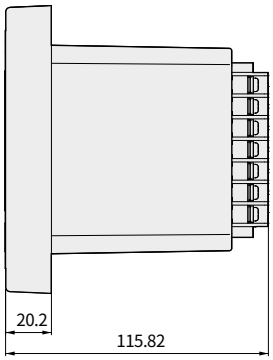
Option Type

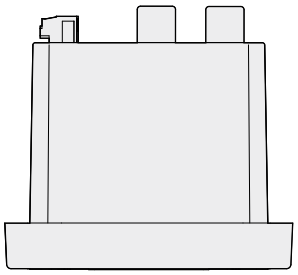
Dimension

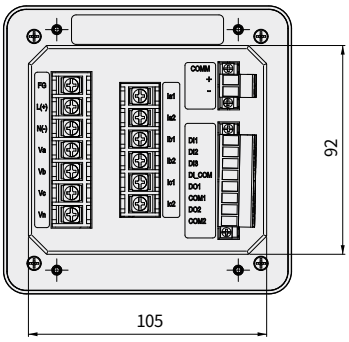
Outline and Panel Cutting

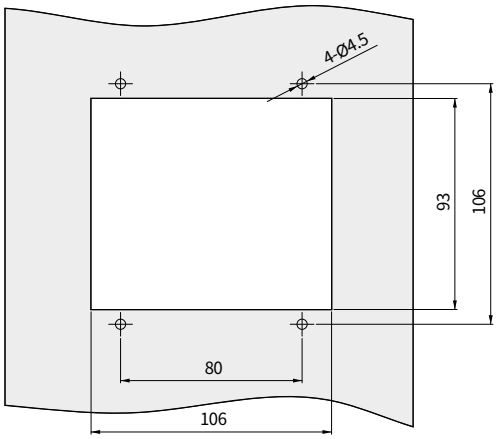
unit: mm / weight: 0.8kg

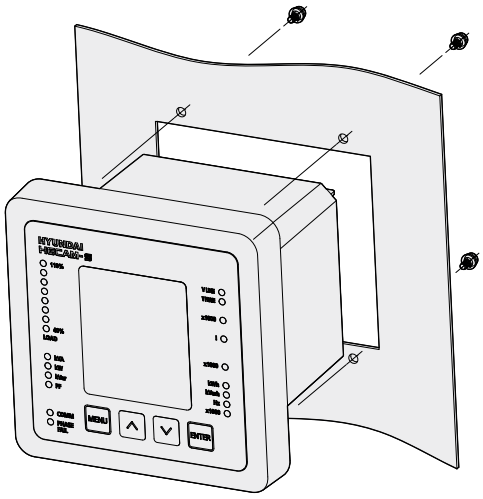






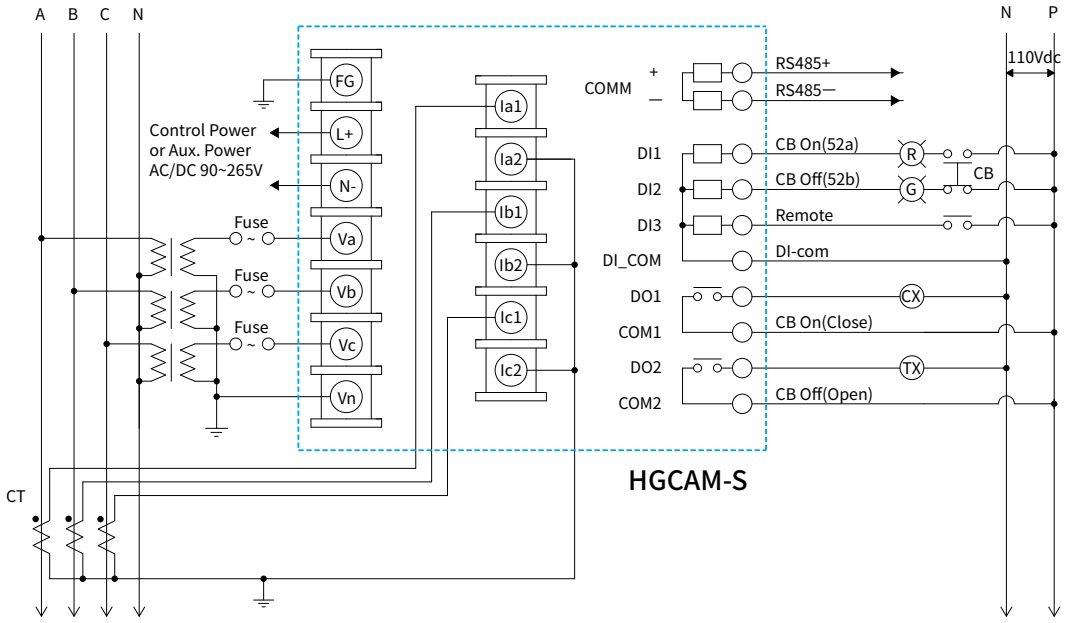






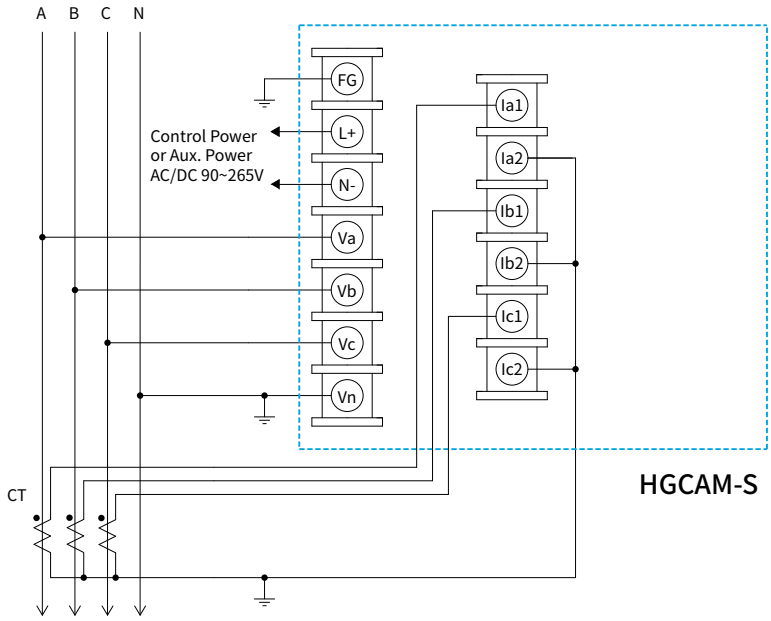
Wiring Diagram

General Wiring Diagram



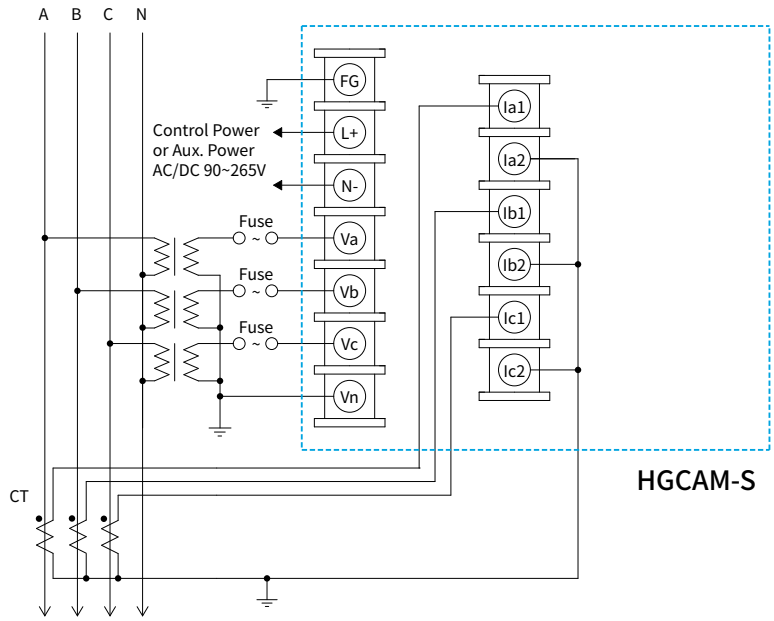
※ Be sure to use the shorting block to connect the meter and CT secondary terminal.
※ The CB ON,OFF relay has a contact rating of 5A at 250VAC, so the auxiliary relay is required to control the circuit breaker.

3P4W Direct



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

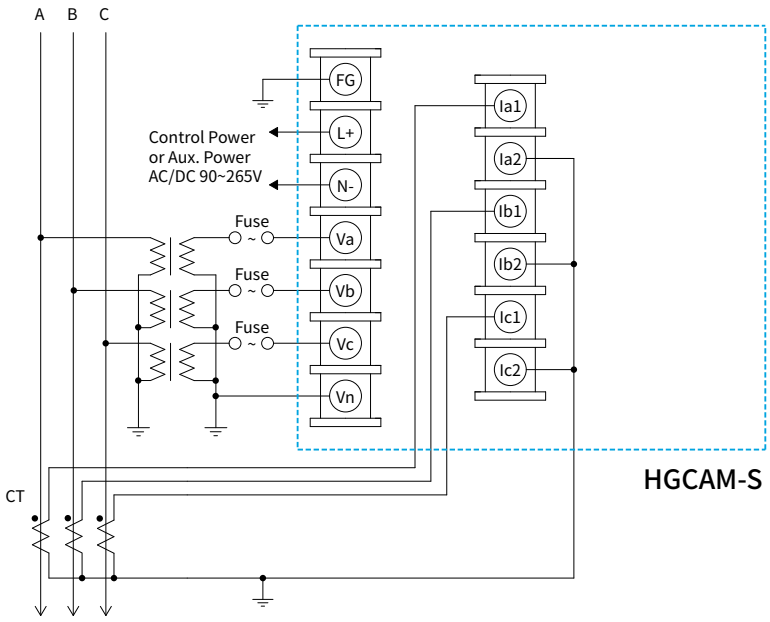
3P4W 3PT/3CT



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

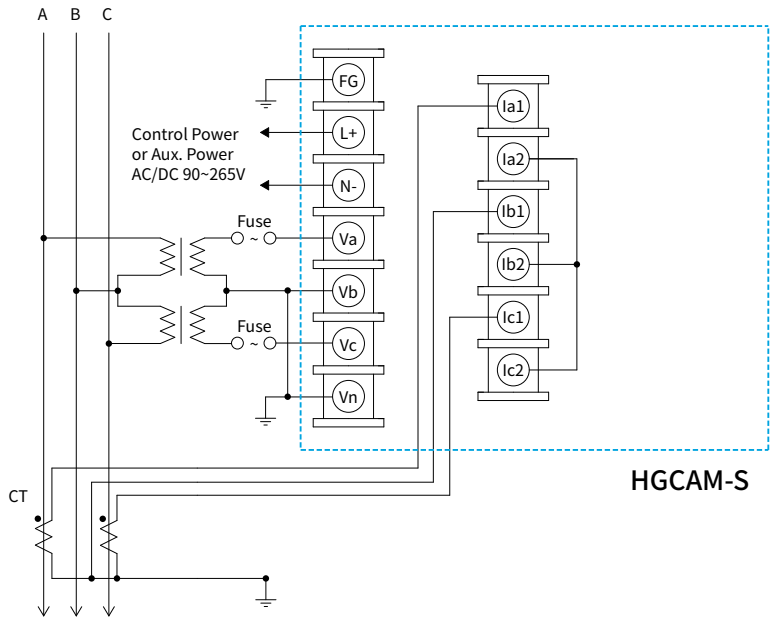
Wiring Diagram

3P3W 3PT/3CT



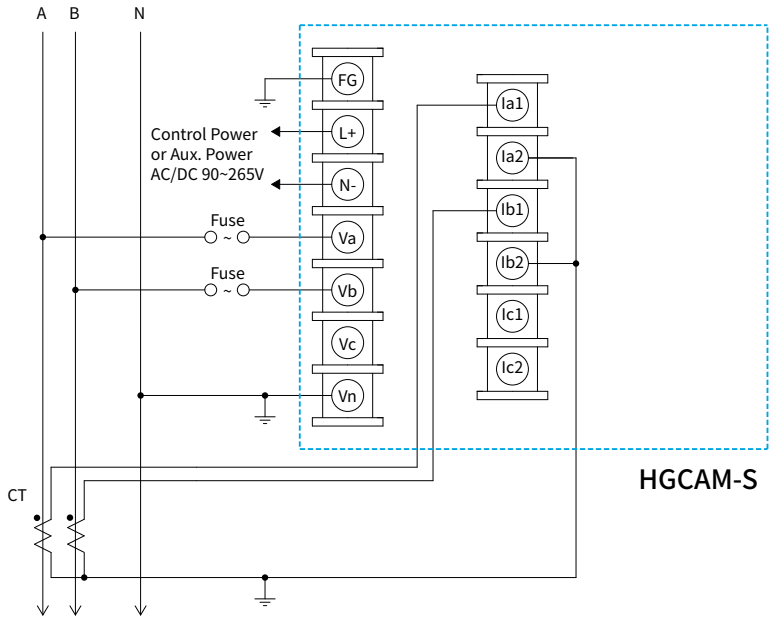
※ Be sure to use the shorting block to connect the meter and CT secondary terminal.
※ Mode setting : 3P4U (3P4W)

3P3W Open Delta



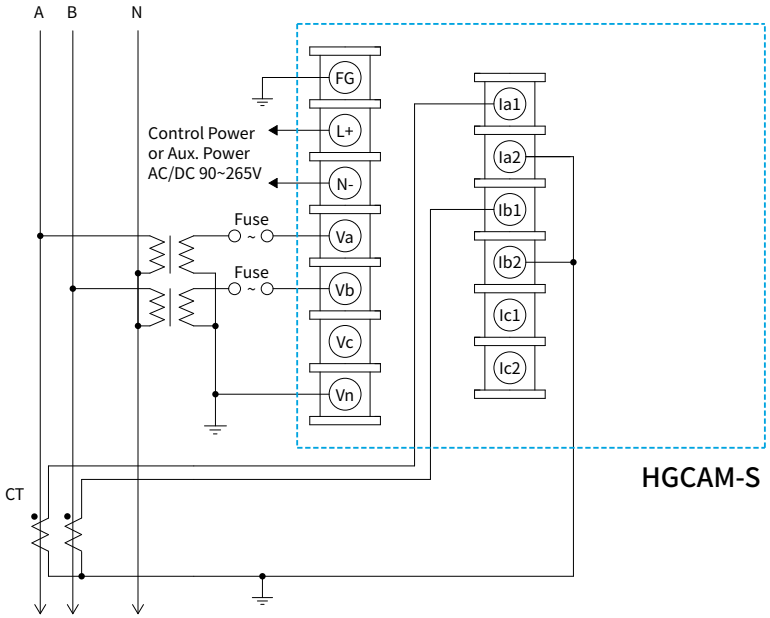
※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

1P3W Direct



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

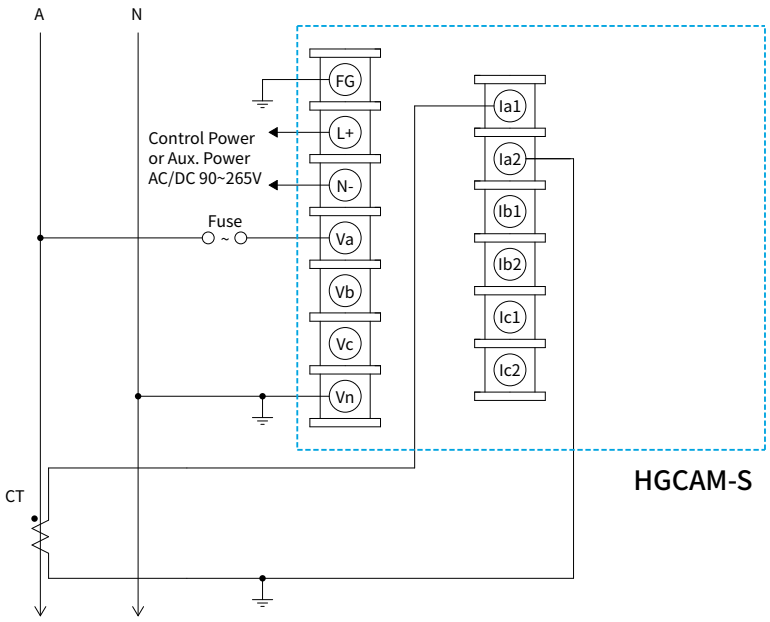
1P3W PT



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

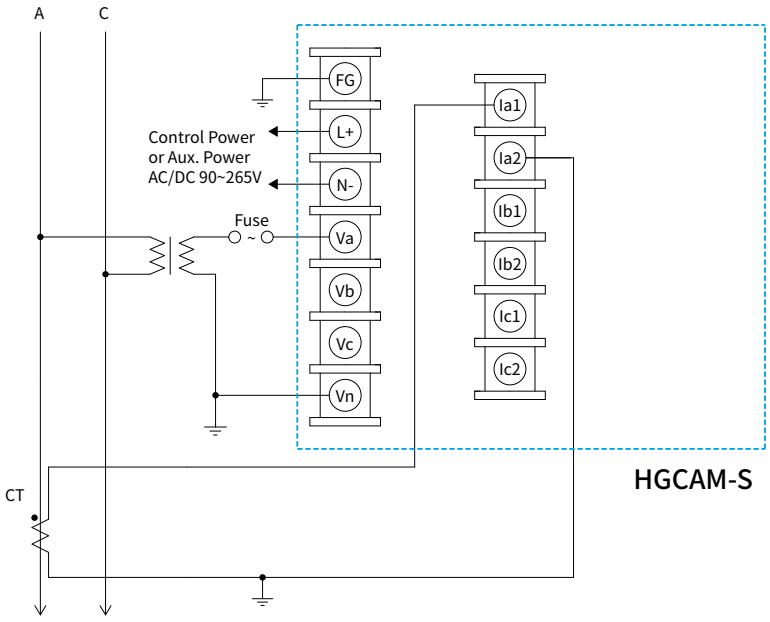
Wiring Diagram

1P2W Direct



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

1P2W PT



※ Be sure to use the shorting block to connect the meter and CT secondary terminal.

Order Code

HGCAM-S

HGCAM	S	N	A5	F6	C1	D1	NO
Category	Type	Outfit	CT 2nd Rating	Frequency	Comm.	I/O Type	Optional
HGCAM Digital Power Meter	**S** Standard type	**N** Rectangular	**A5** AC 5A	**F6** 60Hz	**C1** Modbus-RTU	**D1** DI3DO2 (CB Control)	**NO** No Option
					NO No Option	**NO** No Option	

HG Series

Digital Power DC Meter

HGCAM-S-DC

Compact Size, Strong Performance



Overview

DC Voltage / Current Measurement

Specialized for DC panel to measure DC voltage and current and battery current.

Measurement Accuracy

It satisfies AC/DC voltage ±0.3%, DC current ±0.5%, and can be reliably measured even with frequency fluctuations.

Convenient Installation

Compact design and easy wiring save installation space of panel.

Communication Function

The RS485 MULTI-DROP communication method supports the MODBUS-RTU Protocol.

DI/DO Control

It support 3 channels of DI and 2 channels of DO for convenient connection and system control.

Battery Temperature Measurement

A Thermo-coupler is applied to measure the battery temperature condition for stable operation.



Appearance

Digital Power DC Meter (HGCAM-S-DC)

Front View

No.	Classification	Description
1	FND Display	Display measurement and setting
2	Loadrate LED	Indicate the loadrate based on the rated current
3	Display type LED	ACV : AC voltage DCV : DC voltage DCA : DC current
4	Communication	Communication state
5	AC voltage unit	VAC : AC voltage Hz : Frequency THD : Total harmonic distortion x1000 : 1000x magnification
6	DC voltage unit	VDC : DC voltage x1000 : 1000x magnification
7	Current unit	I Batt : Battery current I DC OUT : DC output current TEMP : (reserved) x1000 : 1000x magnification
8	Control Key	Parameter setting and display mode control



Rear View

Power, PT terminal	
FG	FG
L(+)	Control power
N(-)	
AC1	AC voltage
AC2	
DC+	DC voltage
DC-	

CT terminal	
NC	(reserved)
NC	
OCI+	DC output current
OCI-	
BCI+	Battery current
BCI-	

Comm. terminal	
+	RS 485+
-	RS 485-

DI, DO terminal	
DI1	DI1
DI2	DI2
DI3	DI3
DI_COM	DI_COM
DO1	DO1
COM1	COM1
DO2	DO2
COM2	COM2

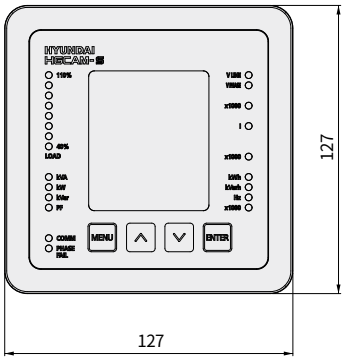


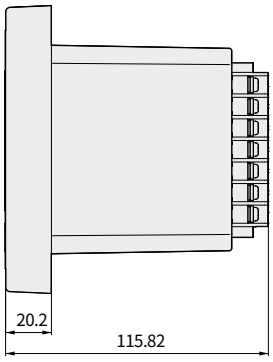
Option Type

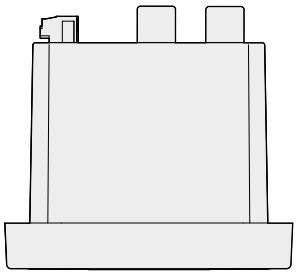
Dimension

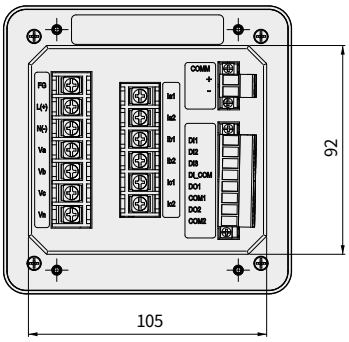
Outline and Panel Cutting

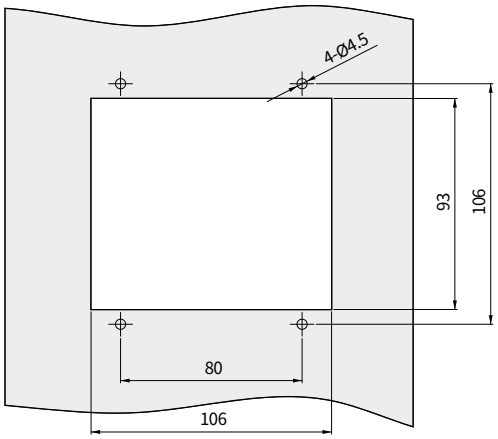
unit: mm / weight: 0.8kg

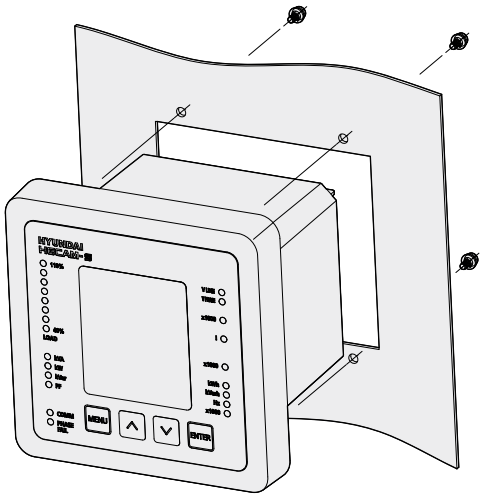




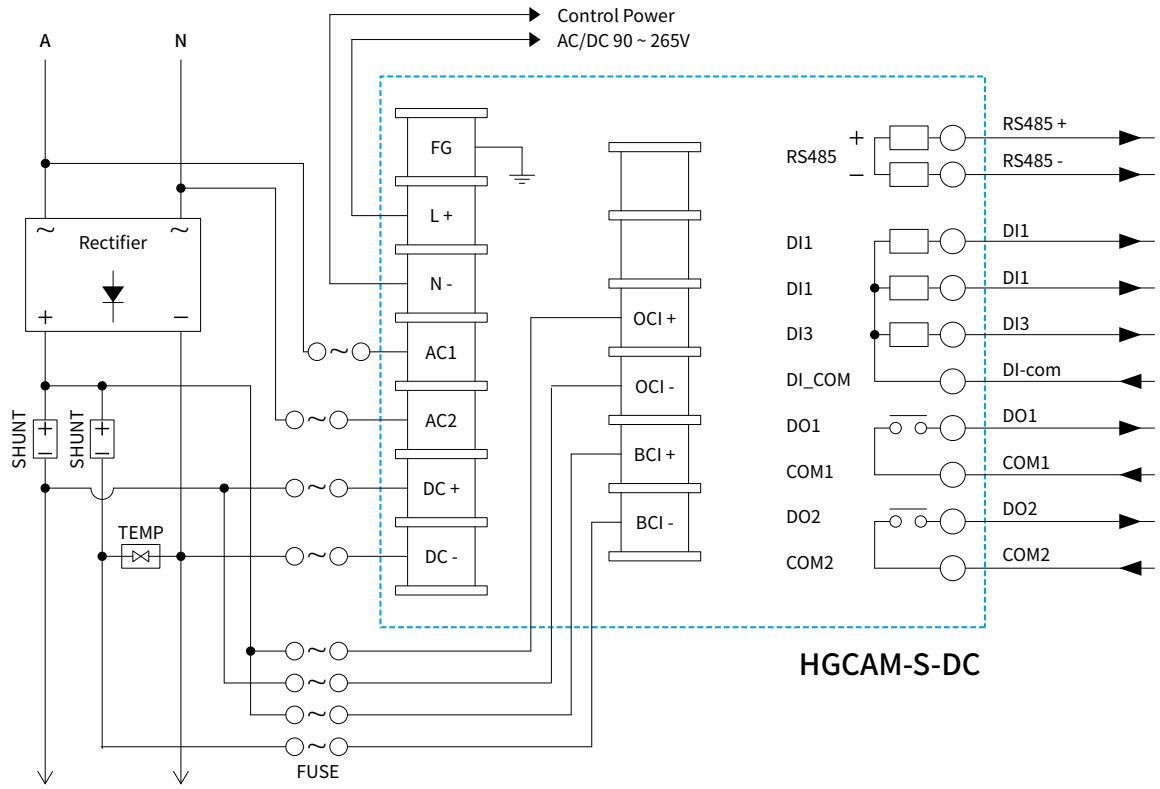








Wiring Diagram



Order Code

HGCAM-S-DC

HGCAM	S	N	DC	F6	C1	D1	NO
Category	Type	Outfit	CT 2nd Rating	Frequency	Comm.	I/O Type	Optional
HGCAM Digital Power Meter	**S** Standard type	**N** Rectangular	**DC** DC	**F6** 60Hz	**C1** Modbus-RTU	**D1** DI3DO2 (CB Control)	**NO** No Option

Operation and Maintenance

Storage and Transportation

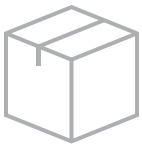
Caution in Transportation

| Temperature | -5~40°C
(average daily temperature below 10°C)
| Location | below 1,000m altitude
| Humidity | 45~85% (relative humidity)

The surrounding environment may affect the insulation performance and durability of the digital power meter, so it must be applied after clearly checking the conditions of the environment.



- **Relative humidity within the range of 45% to 85%**
  - Do not store it in a humid place for a long time.
- **Avoid dusty areas**
  - Do not store in exposed areas, use covers or packaging to prevent dust.



- **Keep it in its original packaging**
  - When storing, keep it in a packaged state.
  - Store on a shelf or similar facility and do not leave it on the floor.
  - Do not carry heavy goods by hand. There is a risk of injury.



- **Do not leave in direct sunlight for a long time**



- **Avoid extremely high or low temperatures**
  - Keep the storage temperature between -20°C and +60°C.



- **Do not store in corrosive gas**
  - Avoid exposure to sulfur gas or ammonia.

Caution in Transportation

⚠ CAUTION

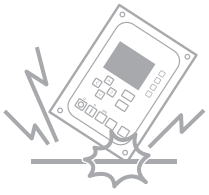
- Do not shock when transporting. Dropping or applying a strong impact can cause defects.
- Do not hold and handle the accessories of the instrument or the external draw-out lines of the accessories. It can cause injuries to the user or malfunction of the instrument.



- **Holding the product body**
  - Do not hold and handle the cable or external lead wire.



- **Care should be taken when handling metal fittings**
  - There are sharp sides or corners, so there is a risk of injury.



- **Be careful of shocks during transport**
  - Dropping or applying a strong impact can cause defects.



- **Pay attention to instrument packaging before transportation**
  - Inadequate packaging can cause damage to the instrument during transportation.

Safety Information

⚠ WARNING

- Ensure that all power source and buses are isolated before installation and wiring work.
- Never attempt to work by oneself.
- Check the connection of the ground terminal before starting operation.
- Do not handle the device with wet hands.
- All cables must be provided with suitable lugs or terminals.
- PT secondary circuit must not be short-circuited.
- CT secondary circuit must not be disconnected.

⚠ CAUTION

- Any installation and maintenance shall be performed by the authorized personnel.
- Check the control voltage condition before wiring.
- Check the rated power and polarity before installation.
- Contaminants such as liquids or bolt, metals inside the device can damage the system.
- Be sure to use the auxiliary relay for circuit breaker.

Precaution during storage

- Store the device in dry and dust-free place away from direct sunlight.
- Make sure that nothing is broken or covered with soot or pollutant.
- Verify that there is no damage to terminal block in the rear.
- Check the correctness of PT/CT/ZCT/GPT rating.

Precaution during installation

- Confirm that the rating of control power meet the specification.
- Ensure that polarity(P,N) is correctly connected to the control power terminal.
- Check that PT, CT secondary is connected well to PT, CT terminal.
- Check that the communication cable is connected to the terminal.
- Be sure to use a shielded twisted pair(STP) cable for communication.

Precaution during test

- Voltage withstand test and insulation resistance test for meter are strictly prohibited. If required, keep separate it from the panel before these tests.
- Circuit shall be separated from CT, PT secondary using CT test terminal and PT test terminal before the test.
- All test equipment shall be get through with proper calibration.
- Check the permissible range and rating for test voltage and current.

Precaution while distribution board design

- Use the auxiliary relay and lockout relay to control the circuit breaker.
- Install the clamping diode when using magnetic contacts and inductive loads in the control power circuit.
- The device FG ground must be connected to the ground conductor which is electrically isolated from the panel enclosure.

Precautions during operation

- Check the negative sequence voltage and current, power factor, active power during operation.
- Do not press the control button without the consent of the site manager.

Precaution during maintenance and replacement

- Follow the detailed work manual.
- Do not replace the device in rainy or humid condition.
- Open PT secondary circuit using the PT test terminal and short CT secondary circuit using CT test terminal.
- Ensure that the relay is clearly opened before maintenance.
- Keep a setting value(PT,CT ratio) in writing before replacement.
- Clear and conspicuous consent of site manager is required for any site work.
- Check that control power is clearly off before replacement.

Certificate

HGCAM-A



9817-A3AF-D14E-9D3C

방송통신기자재등의 적합등록 필증

Registration of Broadcasting and Communication Equipments

상호 또는 성명 Trade Name or Registrant	에이치디현대일렉트릭 주식회사
기자재명칭(제품명칭) Equipment Name	디지털 전력 계측기
기기부호/추가 기기부호 Equipment code /Additional Equipment code	MEA11
기본모델명 Basic Model Number	HGCAM-A
파생모델명 Series Model Number	
등록번호 Registration No.	R-R-00H-HGCAM-A2
제조자/제조국가 Manufacturer/Country of Origin	에이치디현대일렉트릭 주식회사/한국
등록연월일 Date of Registration	2023-10-31
기타 Others	

위 기자재는「전파법」제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2023년(Year) 10월(Month) 31일(Day)

국립전파연구원장

Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
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HGCAM-S



7D29-01B7-3762-EE81

방송통신기자재등의 적합등록 필증

Registration of Broadcasting and Communication Equipments

상호 또는 성명 Trade Name or Registrant	에이치디현대일렉트릭 주식회사
기자재명칭(제품명칭) Equipment Name	디지털 전력 계측기
기기부호/추가 기기부호 Equipment code /Additional Equipment code	MEA11
기본모델명 Basic Model Number	HGCAM-S
파생모델명 Series Model Number	
등록번호 Registration No.	R-R-00H-HGCAM-S2
제조자/제조국가 Manufacturer/Country of Origin	에이치디현대일렉트릭 주식회사/한국
등록연월일 Date of Registration	2023-10-31
기타 Others	

위 기자재는「전파법」제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2023년(Year) 10월(Month) 31일(Day)

국립전파연구원장

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HGCAM-S-DC



1505-7D6E-694A-0C3E

방송통신기자재등의 적합등록증

Registration of Broadcasting and Communication Equipments

상호 또는 성명 Trade Name or Registrant	에이치디현대일렉트릭 주식회사
기자재명칭(제품명칭) Equipment Name	디지털 계측기
기기부호/추가 기기부호 Equipment code /Additional Equipment code	MEA11
기본모델명 Basic Model Number	HGCAM-S-DC
파생모델명 Series Model Number	
등록번호 Registration No.	R-R-00H-HGCAM-S-DC
제조자/제조국가 Manufacturer/Country of Origin	에이치디현대일렉트릭 주식회사/한국
등록연월일 Date of Registration	2020-11-06
기타 Others	

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It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2025년(Year) 09월(Month) 08일(Day)

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HGCAM-PQ



D0A7-699C-DDD3-1855

방송통신기자재등의 적합등록 필증

Registration of Broadcasting and Communication Equipments

상호 또는 성명 Trade Name or Registrant	에이치디현대일렉트릭 주식회사
기자재명칭(제품명칭) Equipment Name	디지털전력품질계측기
기기부호/추가 기기부호 Equipment code /Additional Equipment code	MEA11
기본모델명 Basic Model	HGCAM-PQ
파생모델명 Series Model	
등록번호 Registration No.	R-R-00H-HGCAM-PQ
제조자/제조국가 Manufacturer/Country of	에이치디현대일렉트릭 주식회사/한국
등록연월일 Date of Registration	2023-05-08
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.

It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2023년(Year) 05월(Month) 08일(Day)

국립전파연구원장

Director General of National Radio Research Agency

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Potential for Tomorrow

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Reliability Center	17-10, Mabuk-ro 240beon-gil, Giheung-gu, Yongin-si, Gyeonggi-do, Republic of Korea	Tel: +82-31-289-5000	Fax: +82-31-289-5040
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London	50-68 High Street, Weybridge, KT13 8BL, United Kingdom	Tel: +44-7979-125051	
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Dubai	Unit 205, Emaar Square Building NO.4 Sheikh Zayed Road, Pobox 252458, Dubai, U.A.E	Tel: +971-50-321-8154	Fax: +971-4-425-7996
Bangkok	19th Floor, Unit 1908, Sathorn Square Office Tower, 98 North Sathorn Road, Silom, Bangrak, Bangkok 10500, Thailand	Tel: +66-02-115-7920	Fax: +66-2-115-7898
Ho Chi Minh	Room 1405, 14th Floor Centec Tower, 72-74 Nguyen Thi Minh Khai Street Vo Thi Sau Ward, District 3, Ho Chi Minh City, Vietnam	Tel: +84-28-3535-0465	
Osaka	1-Room 5th Floor Nagahori Plaza Bldg. 2-4-8 Minami Senba, chuo-ku, Osaka 542-0081, Japan	Tel: +81-6-6261-5766~7	Fax: +81-6-6261-5818
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Hungary	H-1145 Bacskai utca 29/a, Budapest, Hungary	Tel: +36-1-273-3733	
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